



On the Formation of Microstructure for Singularly Perturbed Problems with Two, Three, or Four Preferred Gradients

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Abstract

In this manuscript, singularly perturbed energies with 2, 3, or 4 preferred gradients subject to incompatible Dirichlet boundary conditions are studied. This extends results on models for martensitic microstructures in shape memory alloys ($N = 2$), a continuum approximation for the $J_1 - J_3$ -model for discrete spin systems ($N = 4$), and models for crystalline surfaces with N different facets (general N). On a unit square, scaling laws are proved with respect to two parameters, one measuring the transition cost between different preferred gradients and the other measuring the incompatibility of the set of preferred gradients and the boundary conditions. By a change of coordinates, the latter can also be understood as an incompatibility of a variable domain with a fixed set of preferred gradients. Moreover, it is shown how simple building blocks and covering arguments lead to upper bounds on the energy and solutions to the differential inclusion problem on general Lipschitz domains.

Keywords Scaling law · Singular perturbation · Formation of microstructure · N preferred gradients

Mathematics Subject Classification 49J40 · 82B21 · 74N15

1 Introduction

We consider a singularly perturbed energy for 2, 3, or 4 preferred gradients and study the formation of microstructure for incompatible boundary conditions. The energy

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functional under consideration is given by

$$E_{\sigma,\gamma,N}(u) = \int_{(0,1)^2} \text{dist}(\nabla u, K_{\gamma,N})^2 dx + \sigma |D^2 u|((0,1)^2), \quad (1)$$

where $\sigma > 0$, $N \in \{2, 3, 4\}$ and the set of preferred gradients is defined by $K_{\gamma,N} = \left\{ e^{i\gamma + 2i\pi \frac{k}{N}} : k = 0, \dots, N-1 \right\} \subseteq S^1$ for $\gamma \in (-\pi, \pi]$. Our main result concerns a scaling law for the minimal energy subject to incompatible boundary conditions with respect to σ and γ .

Proving scaling laws for the minimal energy has been proved useful to explain the formation of patterns in a variety of problems in which the energy is non-quasiconvex and identifying the minimizers by analytical or numerical methods is not possible, c.f. Kohn (2007). Often the formation of patterns is related to the competition of a part of the energy that favors rather uniform structures and a non-quasiconvex part of the energy that favors oscillations on fine scales. Constructions of good competitors for the upper bound often involve branching construction that refines in a self-similar manner. A non-exhaustive list of references where this technique has been successfully applied includes (Kohn and Müller 1994; Conti 2000, 2006; Capella and Otto 2009, 2012; Chan and Conti 2015; Knüpfer et al. 2013; Bella and Goldman 2015; Conti and Zwicky 2016; Conti et al. 2020; Rüland and Tribuzio 2022) for martensitic microstructure, Kohn and Wirth (2014, 2016) for compliance minimization, Choksi and Kohn (1998), Choksi et al. (1998), DeSimone et al. (2006), Otto and Steiner (2010), Otto and Viehmann (2010), Knüpfer and Muratov (2011), Dabade et al. (2019), Venkatraman et al. (2020); DeSimone et al. (2006) for micromagnetism, Choksi et al. (2004), Choksi et al. (2008), Conti et al. (2016) for type-I-superconductors, Ben Belgacem et al. (2002), Bella and Kohn (2014), Bourne et al. (2017), Conti et al. (2005) for compressed thin elastic films, Conti and Ortiz (2005), Conti and Zwicky 2016) for dislocation patterns, and Ginster and Zwicky 2023), Ginster and Zwicky 2024) for helimagnets.

Energies of the form (1) appear in different contexts. For $N = 2$ and $\gamma = \pi/2$, the energy (1) reads as

$$\int_{(0,1)^2} |\partial_1 u|^2 + \min\{|\partial_2 u + 1|, |\partial_2 u - 1|\}^2 dx + \sigma |D^2 u|((0,1)^2),$$

which is a variant of the energy studied by Kohn and Müller (1994) for pattern formation in shape memory alloys, see also Zwicky 2014), Conti (2006). In particular, in Zwicky 2014) it is shown that

$$\min_{u(0,\cdot)=0} E_{\sigma,\pi/2,2}(u) \sim \min\{1, \sigma^{2/3}\}. \quad (2)$$

Here, the term of order 1 corresponds to rather uniform configurations such as $u = 0$, whereas the bound of order $\sigma^{2/3}$ can be shown by a self-similarly refining branching construction. For more details on the branching pattern, we refer to Fig. 1 and discussion.

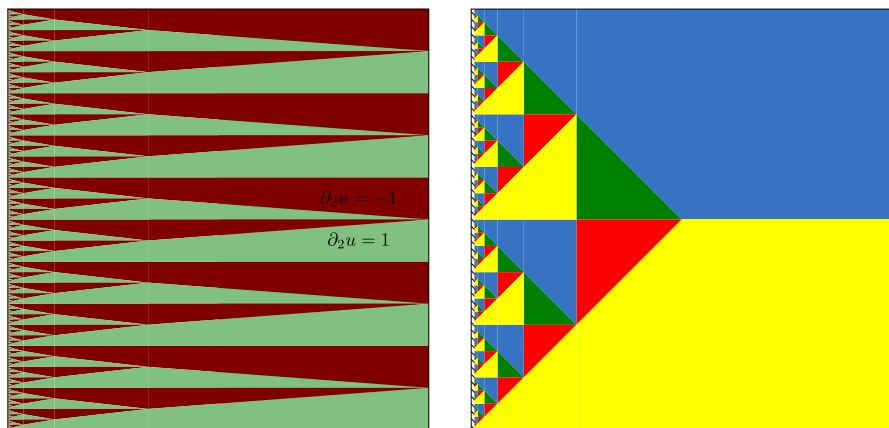


Fig. 1 Sketch of the branching constructions for the energies $E_{\sigma, \pi/2, 2}$ (left) and $E_{\sigma, \pi/4, 4}$ (right). Left: the colors indicate regions where $\partial_2 u = 1$ and $\partial_2 u = -1$. However, $\nabla u \notin K_{\pi/2, 2}$. Right: Up to a small interpolation region, it holds $\nabla u \in K_{\pi/4, 4}$. The colors indicate the regions where the four different preferred gradients from $K_{\pi/4, 4}$ are used: $e^{i\pi/4}$ (blue), $e^{i3\pi/4}$ (red), $e^{i5\pi/4}$ (green), and $e^{i7\pi/4}$ (yellow). Without the interpolation region, this construction would lead to a solution of the differential inclusion $\nabla u \in K_{\gamma/4, 4}$ and $u(0, \cdot) = 0$ (Color figure online)

For $N = 4$ and $\gamma = \pi/4$, the energy (1) is a variant of the energy that appears in certain parameter regimes (after appropriate rescaling) as the continuum approximation of the discrete $J_1 - J_3$ -model

$$F_{\varepsilon, \alpha}(\varphi) = -\alpha \sum_{i, j \in \varepsilon \mathbb{Z}^2 \cap (0, 1)^2: |i-j|=\varepsilon} \varphi(i) \cdot \varphi(j) + \sum_{i, j \in \varepsilon \mathbb{Z}^2 \cap (0, 1)^2: |i-j|=2\varepsilon} \varphi(i) \cdot \varphi(j)$$

associated with a spin field $\varphi : \varepsilon \mathbb{Z}^2 \cap (0, 1)^2 \rightarrow S^1$, see Cicalese et al. (2019), Ginster and Zwischagl (2023) for the heuristic argument and Ginster et al. (2024) for a rigorous derivation via Γ -convergence. A corresponding scaling law is studied in Ginster and Zwischagl (2023) which reads as

$$\min_{u(0, \cdot)=0} E_{\sigma, \pi/4, 4} \sim \min\{1, \sigma(|\log \sigma| + 1)\}. \quad (3)$$

Again, the term of order 1 is associated with rather uniform structures such as $u = 0$, whereas the bound $\sigma(|\log \sigma| + 1)$ corresponds to a (different) self-similarly refining branching construction. For a more details on the branching construction, we refer to discussion and Fig. 1.

For general N , the energy functional (1) is very closely related to the (small slope approximation of the) free energy of a faceted crystalline surface in $\mathbb{R}^3$ with N different facets parameterized by $z = u(x, y)$

$$F_\sigma[u] = \int W(\nabla u) + \frac{\sigma^2}{2} |\Delta u|^2 \, dx,$$

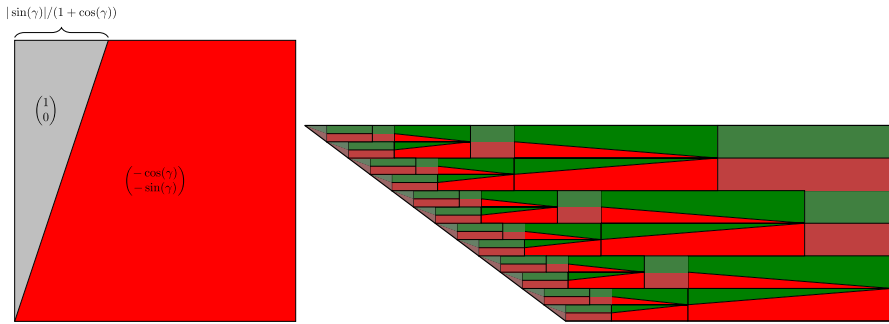


Fig. 2 Left: Sketch of the construction with an energy of order $|\sin(\gamma)|^3 + \sigma$ for $\gamma \leq 0$. Right: Sketch of the modification of the branching construction for $N = 2$ and $\gamma \geq 0$, see Fig. 1, to a specific domain with height $|\sin(\gamma)|$. As in Fig. 1, the colored regions indicate where $\partial_2 u \in \{\pm 1\}$. Note that it holds $u = 0$ on the left boundary. A rotated version of this construction is then used close to $\{0\} \times (0, 1)$, see Fig. 3 (Color figure online)

where W has N distinct minima in S^1 , see, for example, Watson and Norris (2006) and the references therein. Typically, replacing the term $\sigma |D^2 u|^2$ by $\sigma^2 \int_{(0,1)^2} |D^2 u|^2 \, dx dy$ does not qualitatively change corresponding scaling law results, see Zwirnagl (2014) and Schreiber (1994). Hence, up to replacing the Laplacian by a full Hessian it is to be expected that the scaling law results presented in this paper are also valid for F_σ above (c.f. also the discussion in Ginster and Zwirnagl 2023). A study of the dynamics of F_σ , coarsening rates and simulations can, for example, be found in Watson and Norris (2006), Watson (2015).

For general $\gamma \in (-\pi, \pi]$, the energy (1) can by a simple change of variables alternatively be written as

$$\int_{Q_{\gamma_0-\gamma}} \text{dist}(\nabla u, K_{\gamma_0, N})^2 \, dx + \sigma |D^2 u|^2(Q_{\gamma_0-\gamma}),$$

where $Q_{\gamma_0-\gamma}$ evolves from $(0, 1)^2$ by a counterclockwise rotation with angle $\gamma_0 - \gamma$. Hence, our studies can be understood as a generalization of the scaling laws (2) and (3) (cf. Kohn and Müller 1994; Zwirnagl 2014; Ginster and Zwirnagl 2023) either to different domains or to a different set of preferred gradients.

Our main result is the following scaling law

$$\min_{u(0, \cdot)=0} E_{\sigma, \gamma, N}(u) \sim \begin{cases} \min\{|\sin(\gamma)|^2, |\sin(\gamma)|^3 + \sigma, \sigma^{2/3} |\sin(\gamma)|\} & \text{if } N = 2, \\ \min\left\{|\sin(\gamma)|^2, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1\right)\right\} & \text{if } N = 3, 4, \end{cases}$$

where the range of γ is by symmetry considerations restricted to the interval around 0 such that $|\sin(\gamma)| \leq |\sin(\gamma + \frac{2\pi k}{N})|$, $0 \leq k \leq N-1$. In the following, we will comment on different terms appearing in the scaling laws, their connection to the literature, the significant structural differences between the case $N = 2$ and the case $N = 3, 4$, and an outlook to the case $N > 4$.

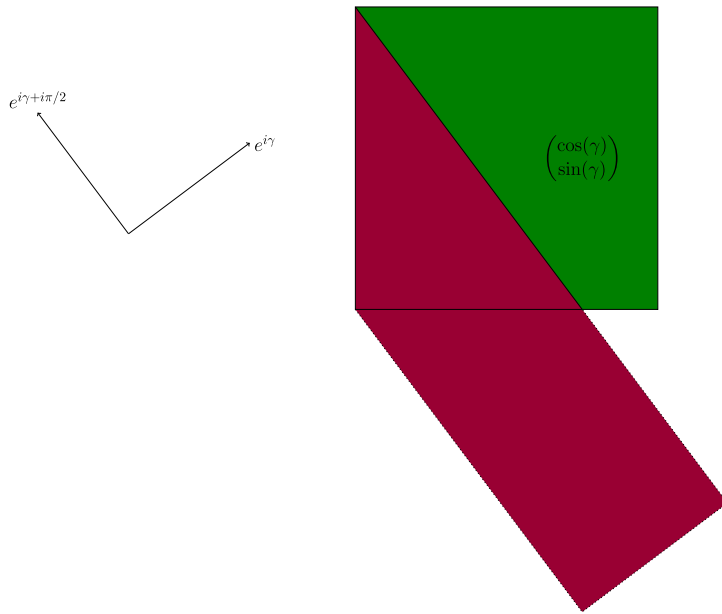


Fig. 3 The purple area indicates the domain in which the modified branching construction from Fig. 2 (right) can be inserted. It can also be used in a similar way for the more complicated constructions for $N = 3$ and $N = 4$, see Proposition 8 Step 1 and Step 2

First, we note that whenever $\gamma \neq 0$ the boundary conditions $u(0, \cdot) = 0$ are incompatible with the set of preferred gradients $K_{\gamma,N}$, i.e., there exists no affine function u with $u(0, \cdot) = 0$ and $\nabla u \in K_{\gamma,N}$. The scaling law quantifies that the boundary conditions and the set of preferred gradients become more and more compatible as $\gamma \rightarrow 0$. Depending on the values of σ and γ , the simplified picture is that in order to meet the boundary values $u(\cdot, y) = 0$ optimal structures are either rather affine with $\partial_2 u \approx 0$ (if σ is relatively large) or ∇u oscillates close to $\{0\} \times (0, 1)$ highly between different gradients such that the y -derivatives average to 0 (for $N = 3, 4$ it is possible to use essentially only gradients in $K_{\gamma,N}$, whereas for $N = 2$ this is not possible, see below). Indeed, in the scaling law above the first term $|\sin(\gamma)|^2$ corresponds for all N to a uniform structure, i.e., $u(x, y) = x$. For $N = 2$, there is an additional term that corresponds to a slightly different but still rather uniform structure, where the function u behaves as $u(x, y) = x$ close to $x = 0$ and has one transition to $u(x, y) = -\cos(\gamma)x - \sin(\gamma)y$ shortly after, see Fig. 2. For the last term, branching structures play a role. However, there are significant differences between the case $N = 2$ and the cases $N = 3, 4$. These can already be understood for the prototypical energies $E_{\sigma,\pi/2,2}$ and $E_{\sigma,\pi/4,4}$, c.f. the scaling laws (2) and (3). The main difference is that for $N = 4$ there exist solutions to the differential inclusion $\nabla u \in K_{\gamma,N}$ and $u(0, \cdot) = 0$ which refine in a self-similar manner toward $\{0\} \times (0, 1)$, see Fig. 1. This solution to the differential inclusion has infinite surface energy but its construction can be terminated after finitely many refinement steps and is then interpolated on a small layer to meet the boundary conditions. After N refinement

steps, the energy of the interpolated function can be bounded (up to a multiplicative constant) by $\sigma N + 2^{-N}$ where essentially the first term corresponds to the created surface energy and the second term is the energy produced by the interpolation. Optimizing in N yields the upper bound $\sigma(|\log \sigma| + 1)$. For general γ , a construction in the same spirit but more complicated can be performed, c.f. Proposition 8, for general γ . In this case, the higher compatibility of the boundary conditions and the set $K_{\gamma,N}$ is expressed by the fact that the construction refines slower toward the boundary and thus creates less surface energy. The heuristic argument is the following. Assuming that $\nabla u \in K_{\gamma,N}$, it holds that $|u(x, y)| \leq x$ since $|\partial_1 u| \leq 1$. Combining this with $\partial_2 u(x, \cdot) \in \{\pm \sin(\gamma), \pm \cos(\gamma)\}$, one expects for small values $|\gamma|$ at least $\sim \frac{|\sin(\gamma)|}{x}$ jumps of $\partial_2 u(x, \cdot)$. The upper bound construction presented in this manuscript in Proposition 8 essentially follows these heuristics and leads to an upper bound of the form $\sigma N + |\sin(\gamma)|^N$ which after optimization in N provides the upper bound from (1).

On the other hand, for $N = 2$ and $\gamma = \pi/2$, it holds $K_{\gamma,N} = \{(0, \pm 1)\}$. Hence, there exists no solution to the differential inclusion $\nabla u \in K_{\gamma,N}$ with $u(\cdot, y) = 0$. Indeed, $\nabla u \in K_{\gamma,N}$ implies that u is constant in the horizontal direction, i.e., $u = 0$ on $(0, 1)^2$. This is incompatible with $\partial_2 u \in \{\pm 1\}$ which is implied by $\nabla u \in K_{\pi/2,2}$. Hence, refining structures will always induce energy through both terms in the energy $E_{\sigma,\pi/2,2}$. Indeed, the celebrated self-similarly refining branching structure by Kohn and Müller (1994), which provides the upper bound $\sigma^{2/3}$, compromises between oscillations of $\partial_2 u \in \{\pm 1\}$ and their surface energy and the energy created by violating $\partial_1 u = 0$, see Fig. 1. For general γ , we modify this branching construction such that it can be used on a triangular portion of the domain close to $\{0\} \times (0, 1)$ with height $|\sin(\gamma)|$, see Figs. 2 and 3. Then the energy $\sigma^{2/3} |\sin(\gamma)|$ corresponds essentially to the scaling $\sigma^{2/3}$ but on a domain of height $|\sin(\gamma)|$ instead of height 1.

Let us note that the energies $E_{\sigma,\pi/2,2}$ and $E_{\sigma,\pi/4,4}$ have also been studied with respect to the boundary conditions $u(\cdot, y) = (1 - 2\theta)y$, $\theta \in (0, 1/2)$, see Kohn and Müller (1994), Zwischagl (2014) and Ginster and Zwischagl (2023). Heuristically, the role of the parameter θ is comparable to the one of γ in the setting here in the sense that for $\theta \rightarrow 0$ the boundary conditions and the set of preferred gradients become more and more compatible. The corresponding shown scaling laws read as

$$\min_{u(\cdot, y) = (1-2\theta)y} E_{\sigma,\pi/2,2} \sim \min\{\theta^2, \sigma^{2/3}\theta^{2/3}\}$$

and

$$\min_{u(\cdot, y) = (1-2\theta)y} E_{\sigma,\pi/4,4} \sim \min\left\{\theta^2, \sigma \left(\frac{|\log \sigma|}{|\log \theta|} + 1\right)\right\}.$$

In the case $N = 2$, the compatibility parameter θ enters the scaling law in a different way than γ which indicates that the two parameters play different roles. For $N = 3, 4$ the scaling laws look similar. Despite this similarity in the cases $N = 3, 4$ on a technical level, the problems are rather different and need different arguments. In particular, a main difficulty in this manuscript is to quantify how the geometry of

$K_{\gamma,N}$ interacts with the geometry of $(0, 1)^2$, whereas in the problem described above the set of preferred gradients is fixed.

Next, it is to be expected that the scaling law (1) is the same for $N > 4$ as for $N = 3, 4$ with constants depending on N . For every fixed N , a lower bound could be established along the lines of the proof for the case $N = 3, 4$. For the upper bounds presented here, it seems that the geometry of the set $K_{\gamma,N}$ has a rather strong influence on the explicit construction. Hence, it appears to be less clear how to construct appropriate competitors for all $N > 4$ in a systematic way. If this can be done in a systematic way, it would in a next step be interesting to study the scaling law of the minimal energy with respect to γ and the number of preferred gradients N .

Additionally, for $N = 3, 4$ we discuss how upper bounds can be constructed using simple building blocks and covering arguments. More precisely, the structures of $K_{\gamma,N}$ allow us to construct functions $u : T \rightarrow \mathbb{R}$ (where T is a (rotated) square if $N = 4$ or a particular triangle for $N = 3$) such that $u = 0$ on ∂T , $\nabla u \in K_{\gamma,N}$ a.e. and $|D^2 u|(T) \leq C$ (see Fig. 10), c.f. also Conti et al. (2017), Růland et al. (2018), Cesana et al. (2020), Conti (2008), Cesana et al. (2014), Pompe (2010), Müller and Sverak (1999) and the references therein for constructions in the significantly more complicated vectorial setting. Then simple covering arguments allow us to construct upper bounds on more general domains and give rise to solutions of the differential inclusion $u = 0$ on $\partial\Omega$ and $\nabla u \in K_{\gamma,N}$. Similar to Růland et al. (2018) (c.f. also Ginster and Zwischagl, 2023), by interpolation inequalities regularity of these solutions can be established in certain fractional Sobolev spaces.

In the following, we will fix notation and state the precise model under consideration. In Sect. 3, we will present our main results and discuss the organization of the proofs.

2 Notation and Setting of the Problem

We will write C or c for generic constants that may change from line to line but do not depend on the problem parameters. We write $\log$ to denote the natural logarithm. We write $A \sim B$ for $A, B \in \mathbb{R}$ if there exists a universal constant $C > 0$ such that $\frac{1}{C}A \leq B \leq CA$. For the ease of notation, we always identify vectors with their transposes. Moreover, we will identify $\mathbb{C}$ with $\mathbb{R}^2$ and denote by e_1 and e_2 the two canonical basis vectors for $\mathbb{R}^2$.

For a measurable set $B \subseteq \mathbb{R}^n$ with $n = 1, 2$, we use the notation $|B|$ or $\mathcal{L}^n(B)$ to denote its n -dimensional Lebesgue measure. In addition, for $B \subseteq \mathbb{R}^2$ we write $\text{conv}(B) \subseteq \mathbb{R}^2$ for its convex hull and $\text{int}(B)$ for its interior.

For $\gamma \in [-\pi, \pi]$ and $N \in \{2, 3, 4\}$ we set

$$K_{\gamma,N} = \left\{ e^{i\gamma + 2i\pi \frac{l}{N}} : 0 \leq l \leq N-1 \right\} \subseteq \mathbb{R}^2.$$

The set of admissible functions is defined as

$$\mathcal{A} := \left\{ u \in W^{1,2}((0, 1)^2) : \nabla u \in BV((0, 1)^2), u(0, \cdot) = 0 \right\}$$

For $\sigma > 0$, $N \in \{2, 3, 4\}$ and $\gamma \in [-\pi, \pi]$, we consider the functional $E_{\sigma, \gamma, N} : \mathcal{A} \rightarrow [0, \infty)$ by

$$E_{\sigma, \gamma, N}(u) = \int_{(0,1)^2} \text{dist}^2(\nabla u, K_{\gamma, N}) \, dx + \sigma |D^2 u|((0, 1)^2).$$

The expression $|D^2 u|(\Omega)$ in the second term of the functional $E_{\sigma, \gamma, N}$ denotes the total variation of the vector measure $D^2 u$. By symmetry considerations, it will be enough to consider the angles $\gamma \in [-\pi/2, \pi/2]$ such that $|e^{i\gamma} \cdot e_2| = |\sin(\gamma)|$ is minimal within the set $K_{\gamma, N}$ i.e., $\gamma \in \Gamma_N$, where $\Gamma_N = [-\pi/2, \pi/2]$ for $N = 2$, $\Gamma_N = [-\pi/6, \pi/6]$ for $N = 3$, and $\Gamma_N = [-\pi/4, \pi/4]$ for $N = 4$.

Note that $u \in \mathcal{A}$ in particular implies that $u \in W^{1,1}((0, 1)^2)$ and $\nabla u \in BV$. Hence, u has a continuous representative on the closed square $[0, 1]^2$, see, e.g., Conti and Ortiz (2016, Lemma 9). We will always identify such functions with their continuous representatives.

For an open set $B \subseteq \mathbb{R}^2$ and $u \in W^{1,2}(B)$ with $\nabla u \in BV(B)$, we use the notation $E_{\sigma, \gamma, N}(u; B)$ for the energy on B , i.e.,

$$E_{\sigma, \gamma, N}(u; B) = \int_B \text{dist}^2(\nabla u, K_{\gamma, N}) \, dx + \sigma |D^2 u|(B).$$

In addition for $x \in (0, 1)$, $I \subseteq (0, 1)$ and $u \in \mathcal{A}$ we write

$$E_{\sigma, \gamma, N}(u; \{x\} \times I) = \int_I \text{dist}^2(\nabla u(x, y), K_{\gamma, N}) \, dy + \sigma |\partial_2 \nabla u(x, \cdot)|(I).$$

Note that since $\nabla u \in BV((0, 1)^2)$ this formula makes sense for almost every $x \in (0, 1)$ in the sense of slicing of BV -functions, see Ambrosio et al. (2000). Similarly, we write for $y \in (0, 1)$ and $u \in \mathcal{A}$

$$E_{\sigma, \gamma, N}(u; I \times \{y\}) = \int_I \text{dist}^2(\nabla u(x, y), K_{\gamma, N}) \, dx + \sigma |\partial_1 \nabla u(\cdot, y)|(I).$$

Eventually, we define for $B \subseteq (0, 1)^2$ open with Lipschitz boundary the set

$$\mathcal{A}_0(B) := \left\{ u \in W^{1,2}(B) : \nabla u \in BV(B), u = 0 \text{ on } \partial B \right\}.$$

3 Main Result

Our main result is the following scaling law for the minimal energy.

Theorem 1 *There exist constants $C, c > 0$ such that for all $\sigma > 0$ and $\gamma \in \Gamma_N$ it holds:*

1. If $N = 2$, then

$$c \min \left\{ \sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2 \right\} \\ \leq \min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \leq C \min \left\{ \sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2 \right\}.$$

2. If $N = 3$ or $N = 4$, then

$$c \min \left\{ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right), |\sin(\gamma)|^2 \right\} \\ \leq \min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \leq C \min \left\{ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right), |\sin(\gamma)|^2 \right\}.$$

The proof of the theorem is split into different sections. Bounds that are valid for all $N \in \{2, 3, 4\}$ are collected in Sect. 4. Specific upper and lower bounds for $N = 2$ are proved in Sect. 5. The lower bound for $N = 3, 4$ can be found in Proposition 7 in Sect. 6. The corresponding upper bound is shown in Proposition 8 in Sect. 6.

Below, we identify different regimes that appear in the bounds in Theorem 1.

Remark 1 Note that all claimed bounds from Theorem 1 are actually relevant. In fact, it holds for $\gamma \in \Gamma_2$ that

$$\min \left\{ \sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2 \right\} \\ \sim \begin{cases} |\sin(\gamma)|^2 & \text{if } \sigma \geq |\sin(\gamma)|^2, \\ \sigma & \text{if } |\sin(\gamma)|^2 \geq \sigma \geq |\sin(\gamma)|^3, \\ |\sin(\gamma)| \sigma^{2/3} & \text{if } |\sin(\gamma)|^3 \geq \sigma. \end{cases}$$

For $N = 3, 4$ and $\gamma \in \Gamma_N$, it holds

$$\min \left\{ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right), |\sin(\gamma)|^2 \right\} \sim \begin{cases} |\sin(\gamma)|^2 & \text{if } \sigma \geq |\sin(\gamma)|^2, \\ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right) & \text{if } \sigma \leq |\sin(\gamma)|^2. \end{cases}$$

A proof of the statements above is found in appendix.

Additionally, we also consider for $N = 3, 4$ the minimization problem subject to $u = 0$ on $\partial(0, 1)^2$. In this situation, the scaling laws for $N = 3$ or $N = 4$ differ due to different incompatibilities of $K_{\gamma, N}$ with respect to the full boundary of $(0, 1)^2$.

Theorem 2 There exist constants $C, c > 0$ such that it holds for all $\gamma \in \Gamma_4$

$$c \min \left\{ 1, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right) \right\} \\ \leq \min_{u \in \mathcal{A}_0((0, 1)^2)} E_{\sigma, \gamma, 4}(u) \leq C \min \left\{ 1, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right) \right\}.$$

Moreover, it holds for all $\gamma \in \Gamma_3$

$$c \min \{1, \sigma (|\log \sigma| + 1)\} \leq \min_{u \in \mathcal{A}_0} E_{\sigma, \gamma, 3}(u) \leq C \min \{1, \sigma (|\log \sigma| + 1)\}.$$

Upper bounds can be shown by means of optimal coverings with suitable building blocks. The same arguments lead to upper bounds and solutions to the differential inclusion problem on general Lipschitz domains.

Proposition 3 *Let $N = 3, 4$ and $\Omega \subseteq \mathbb{R}^2$ a Lipschitz domain. Then the following hold:*

1. *There exists a constant $C_\Omega > 0$ such that*

$$\min_{u \in \mathcal{A}_0(\Omega)} E_{\sigma, \gamma, N}(u; \Omega) \leq C_\Omega \sigma (|\log \sigma| + 1).$$

2. *There exists $u \in W_0^{1, \infty}(\Omega)$ such that $\nabla u \in BV_{loc}(\Omega; K_{\gamma, N})$ and $\nabla u \in W^{s, q}$ for all $0 < s < 1$, $q \in (1, \infty)$ satisfying $\frac{1}{q} > s$.*

The proofs are discussed in Sect. 7.

4 Preliminaries

We first show the energy bounds that are true for all $N \in \{2, 3, 4\}$. The argument in the lower bound is closely related to the ones in Ginster and Zwicknagl (2023, Lemma 5) (c.f. also Ginster and Zwicknagl, 2024).

Proposition 4 *There exists $c_A > 0$ such that for all $\gamma \in \Gamma_N$, $N \in \{2, 3, 4\}$, it holds*

$$c_A \min \{|\sin(\gamma)|^2, \sigma\} \leq \min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \leq |\sin(\gamma)|^2.$$

Proof *Step 1: Upper bound.* Consider $u : (0, 1)^2 \rightarrow \mathbb{R}$ defined by $u(x, y) = \cos(\gamma)x$. Then

$$E_{\sigma, \gamma, N}(u) \leq |\cos(\gamma)e_1 - e^{i\gamma}|^2 = |\sin(\gamma)|^2.$$

Step 2: Lower bounds. Let $u \in \mathcal{A}$. Without loss of generality, we may assume that it holds $E_{\sigma, \gamma, N}(u) \leq \frac{1}{48} \min \{|\sin(\gamma)|^2, \sigma\}$ (otherwise there is nothing to show). By a standard slicing and Fubini-type argument, we find $\bar{x}, y_1, y_2 \in (0, 1)$ such that $y_2 - y_1 \geq 1/2$,

$$\int_0^1 \text{dist}(\nabla u(s, y_i), K_{\gamma, N})^2 ds + \sigma |\partial_1 \nabla u|((0, 1) \times \{y_i\}) \leq 4E_{\sigma, \gamma, N}(u)$$

and

$$\int_0^1 \text{dist}(\nabla u(\bar{x}, t), K_{\gamma, N})^2 dt + \sigma |\partial_2 \nabla u|(\{\bar{x}\} \times (0, 1)) \leq E_{\sigma, \gamma, N}(u).$$

In particular, there exists $t \in (0, 1)$ and $\xi \in K_{\gamma, N}$ such that t is a point of approximate continuity of $\nabla u(\cdot, y_1)$ and it holds $|\nabla u(t, y_1) - \xi| \leq \frac{2}{\sqrt{48}} |\sin(\gamma)|$. Additionally, it holds

$$\begin{aligned} & |\partial_1 \nabla u|((0, 1) \times \{y_1\}), |\partial_1 \nabla u|((0, 1) \times \{y_2\}), |\partial_2 \nabla u|(\{\bar{x}\} \times (0, 1)) \\ & \leq \frac{4}{\sigma} E_{\sigma, \gamma, N}(u). \end{aligned}$$

Since $\frac{4}{\sigma} E_{\sigma, \gamma, N}(u) \leq \frac{1}{12}$ it follows for a.e. $s \in (0, 1)$

$$\begin{aligned} & |\nabla u(s, y_1) - \xi| \leq |\nabla u(t, y_1) - \xi| + |\partial_1 \nabla u|((0, 1) \times \{y_1\}) \\ & \leq \frac{1}{\sqrt{12}} |\sin(\gamma)| + \frac{1}{12} \leq \frac{1}{\sqrt{12}} + \frac{1}{12} \leq \frac{1}{\sqrt{2}}. \end{aligned}$$

Similarly, we estimate for a.e. $s \in (0, 1)$

$$\begin{aligned} & \max\{|\nabla u(s, y_2) - \xi|, |\nabla u(\bar{x}, s) - \xi|\} \\ & \leq |\nabla u(t, y_1) - \xi| + |\partial_1 \nabla u|((0, 1) \times \{y_1\}) + |\partial_1 \nabla u|((0, 1) \times \{y_2\}) \\ & \quad + |\partial_2 \nabla u|(\{\bar{x}\} \times (0, 1)) \leq \frac{1}{\sqrt{12}} |\sin(\gamma)| + \frac{3}{12} \leq \frac{7}{12} \leq \frac{1}{\sqrt{2}}. \end{aligned}$$

Since the distance between different points in $K_{\gamma, N}$ is at least $\sqrt{2}$, it follows for almost all $s \in (0, 1)$ that

$$\begin{aligned} & |\nabla u(s, y_1) - \xi| = \text{dist}(\nabla u(s, y_1), K_{\gamma, N}), \\ & |\nabla u(s, y_2) - \xi| = \text{dist}(\nabla u(s, y_2), K_{\gamma, N}), \\ & \text{and } |\nabla u(\bar{x}, s) - \xi| = \text{dist}(\nabla u(\bar{x}, s), K_{\gamma, N}). \end{aligned}$$

Then we estimate using $u(0, \cdot) = 0$

$$\begin{aligned} & |u(\bar{x}, y_2) - u(\bar{x}, y_1)| \\ & \leq \int_0^{\bar{x}} |\partial_1 u(s, y_2) - \xi_1| + |\partial_1 u(s, y_1) - \xi_1| \, ds \\ & \leq \int_0^1 \text{dist}(\nabla u(s, y_2), K_{\gamma, N}) + \text{dist}(\nabla u(s, y_1), K_{\gamma, N}) \, ds \\ & \leq \left(\int_0^1 \text{dist}(\nabla u(s, y_2), K_{\gamma, N})^2 \, ds \right)^{1/2} + \left(\int_0^1 \text{dist}(\nabla u(s, y_1), K_{\gamma, N})^2 \, ds \right)^{1/2} \\ & \leq 4\sqrt{E_{\sigma, \gamma, N}(u)}. \end{aligned}$$

On the other hand,

$$|u(\bar{x}, y_2) - u(\bar{x}, y_1)| = \left| \int_{y_1}^{y_2} \partial_2 u(\bar{x}, s) - \xi_2 + \xi_2 \, ds \right|$$

$$\begin{aligned}
&\geq \frac{1}{2} |\xi_2| - \int_0^1 \text{dist}(\nabla u(\bar{x}, s), K_{\gamma, N}) \, ds \\
&\geq \frac{1}{2} |\sin(\gamma)| - \left(\int_0^1 \text{dist}(\nabla u(\bar{x}, s), K_{\gamma, N})^2 \, ds \right)^{1/2} \\
&\geq \frac{1}{2} |\sin(\gamma)| - \frac{1}{\sqrt{48}} |\sin(\gamma)| \geq \frac{1}{4} |\sin(\gamma)|,
\end{aligned}$$

where we used the definition of Γ_N for the second inequality. Consequently, we find

$$E_{\sigma, \gamma, N}(u) \geq \frac{1}{16^2} |\sin(\gamma)|^2 \geq \frac{1}{16^2} \min\{\sigma, |\sin(\gamma)|^2\}.$$

□

5 The Problem with Two Gradients

In this section, we prove the scaling law for $N = 2$ claimed in Theorem 1. As explained in introduction, the additional upper bound in the regime $\sigma \leq |\sin(\gamma)|^3$ is proved using a variant of the celebrated Kohn–Müller branching construction developed in Kohn and Müller (1994), c.f. also Conti et al. (2017), Zwischagl (2014). The lower bound uses a variant of the argument from Conti (2006), c.f. also Conti et al. (2017), Zwischagl (2014). The original argument from Conti (2006) is based on identifying the length scale on which one expects to find a jump of $\partial_y u$ on a vertical slice. If a jump of $\partial_y u$ is present, then the surface term of the energy yields a lower bound. If no jump of $\partial_2 u$ can be found, then the lower bound can be proved from the first term of the energy. In the setting of this manuscript, this argument has to be modified to incorporate the geometry of $K_{\gamma, N}$. In particular, the appearing length scale is different and the vertical slices have to be replaced by lines parallel to $e^{i\gamma}$.

Proof of Theorem 1 for $N = 2$. Step 1: Upper bounds.

First assume $\gamma \in [-\pi/2, 0]$ ($\gamma \in (0, \pi/2]$ can be treated analogously) and consider the function $u : \Omega \rightarrow \mathbb{R}$ defined by (see also Fig. 2)

$$u(x, y) = \begin{cases} x & \text{if } (x, y) \cdot v \leq 0, \\ e^{i\gamma + i\pi} \cdot (x, y) & \text{if } (x, y) \cdot v \geq 0, \end{cases}$$

where $v = (-\sin(\gamma), 1 + \cos(\gamma))$. Note that

$$|e_1 - e^{i\gamma}|^2 = |1 - \cos(\gamma)|^2 + |\sin(\gamma)|^2 \leq C |\gamma|^4 + |\sin(\gamma)|^2 \leq C |\sin(\gamma)|^2.$$

Moreover, it holds that $\mathcal{L}^2(\{(x, y) \in (0, 1)^2 : (x, y) \cdot v \leq 0\}) \leq \frac{|\sin(\gamma)|}{1 + \cos(\gamma)} \leq |\sin(\gamma)|$. Consequently, we obtain

$$E_{\sigma, \gamma, 2}(u) = \int_{(0, 1)^2} \text{dist}(\nabla u, K_{\gamma, 2})^2 \, dx + \sigma |D^2 u|((0, 1)^2) \leq C(|\sin(\gamma)|^3 + \sigma).$$

By Remark 1 and the upper bound in Proposition 4, it remains to show that there exists $u : (0, 1)^2 \rightarrow \mathbb{R}$ such that $E_{\sigma, \gamma, 2}(u) \leq C |\sin(\gamma)| \sigma^{2/3}$ if $\sigma \leq |\sin(\gamma)|^3$. In the following, we will assume $0 \leq \gamma \leq \frac{\pi}{2}$ ($\gamma < 0$ can be treated similarly). We first present a modified version of the self-similar Kohn–Müller branching construction, see Kohn and Müller (1994), Zwischagl (2014), Conti et al. (2017), on the domain (see Fig. 5 for a sketch)

$$S_\gamma = \text{conv} \left\{ \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \begin{pmatrix} 1 \\ \sin(\gamma) \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\},$$

i.e., we will construct a function $u : S_\gamma \rightarrow \mathbb{R}$ such that $u = 0$ on the line $\text{conv} \left\{ \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$ and $E_{\sigma, \pi/2, 2}(u; S_\gamma) \leq C |\sin(\gamma)| \sigma^{2/3}$. We will need two slightly different constructions for $0 \leq \gamma \leq \frac{\pi}{4}$ and $\frac{\pi}{4} \leq \gamma \leq \frac{\pi}{2}$, respectively, see Fig. 5. As in the original construction, we fix for the proof some $\theta \in (\frac{1}{4}, \frac{1}{2})$.

Step 1a: Branching on S_γ for $\gamma \in (0, \pi/4]$:

Consider the building block for the branching construction $u_{bb} : (\theta, 1) \times \mathbb{R} \rightarrow \mathbb{R}$ defined for $(x, y) \in (\theta, 1) \times (0, \sin(\gamma))$ as (see Fig. 5)

$$u_{bb}(x, y) = \begin{cases} y & \text{if } 0 \leq y \leq \frac{\sin(\gamma)}{4} + \sin(\gamma) \frac{x-\theta}{4(1-\theta)}, \\ -y + \sin(\gamma) \frac{x-\theta}{2(1-\theta)} + \frac{\sin(\gamma)}{2} & \text{if } \frac{\sin(\gamma)}{4} + \sin(\gamma) \frac{x-\theta}{4(1-\theta)} \leq y \leq \frac{\sin(\gamma)}{2}, \\ y + \sin(\gamma) \frac{x-\theta}{2(1-\theta)} - \frac{\sin(\gamma)}{2} & \text{if } \frac{\sin(\gamma)}{2} \leq y \leq \frac{3\sin(\gamma)}{4} - \sin(\gamma) \frac{x-\theta}{4(1-\theta)}, \\ -y + \sin(\gamma) & \text{if } \frac{3\sin(\gamma)}{4} - \sin(\gamma) \frac{x-\theta}{4(1-\theta)} \leq y \leq \sin(\gamma). \end{cases}$$

and extended $\sin(\gamma)$ -periodically in y . Note that $u(\theta, y) = \frac{1}{2}u(1, 2y)$ and $u(\cdot, 0) = u(\cdot, \sin(\gamma)) = 0$.

Additionally, define the function $u_\partial : \text{conv} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \begin{pmatrix} 0 \\ \sin(\gamma) \end{pmatrix} \right\} =: \Delta_\gamma \rightarrow \mathbb{R}$ as (see Fig. 4)

$$u_\partial(x, y) = \begin{cases} \frac{\sin(\gamma)}{\cos(\gamma)}x + y & \text{if } y \leq \sin(\gamma) - \frac{\sin(\gamma)}{2\cos(\gamma)}(x + \cos(\gamma)), \\ -y + \sin(\gamma) & \text{if } y \geq \sin(\gamma) - \frac{\sin(\gamma)}{2\cos(\gamma)}(x + \cos(\gamma)). \end{cases}$$

Now, let $K, M \in \mathbb{N}$ be fixed. Then we define the function $u_K : S_\gamma \rightarrow \mathbb{R}$ as described below (see Fig. 4). Here, the parameter M determines the number of jumps of $\partial_2 u_K$ on the right of the domain S_γ , whereas K corresponds to the number of refinement steps of the branching construction.

1. Let $0 \leq m \leq M - 1$. Then define for $(x, y) \in (1 - \frac{m}{M}\cos(\gamma), 1) \times (\frac{m\sin(\gamma)}{M}, \frac{(m+1)\sin(\gamma)}{M})$

$$u_K(x, y) = \frac{1}{M}u_{bb}(1, My).$$

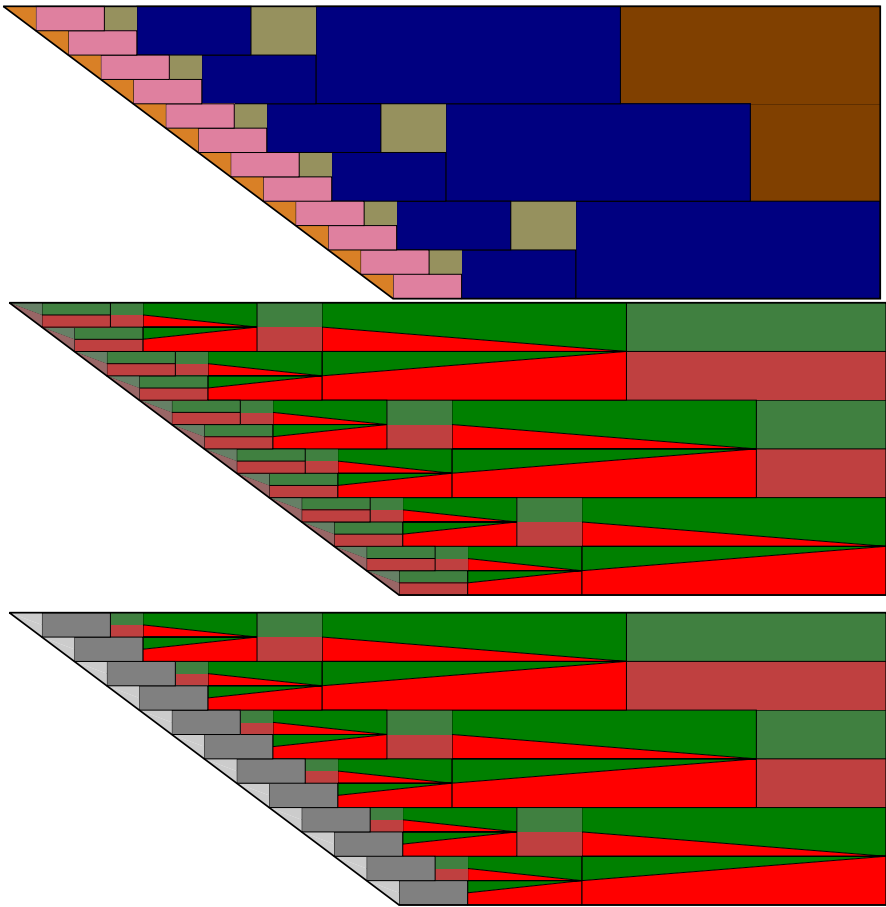


Fig. 4 Sketch of the Kohn-Müller type branching constructions. Top: sketch of the regions appearing in the definition of u_K in (1) (brown), (2) (blue), (3) (beige), (4) (pink), and (5) (orange) on the domain S_y . In the regions (1) and (3), the function is constant in x -direction. Middle: sketch for small y . The interpolation between the boundary values and the branching construction uses an extension to region (4) which is constant in x and the function $u\delta$ as a building block in region (5). The regions with $\partial 2uK = 1$ are colored in red; the region with $\partial 2uK = -1$ is colored in green. Bottom: sketch for large y . In region (4), a linear interpolation in x to $uK = 0$ is used. Then uK is the extended by 0 into region (5). The regions with $\partial 2uK = 1$ are colored in red; the region with $\partial 2uK = -1$ is colored in green (Color figure online)

2. Let $0 \leq k \leq K, 0 \leq m \leq M - 1, a \in \{(a_1, \dots, a_k) : a_i \in \{0, 1\}, 1 \leq i \leq k\}$ and

$$(x, y) \in \left(\theta^{k+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M}, \theta^k - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \\ \times \left(\frac{m}{M} \sin(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M}, \frac{m}{M} \sin(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-k} \frac{\sin(\gamma)}{M} \right).$$

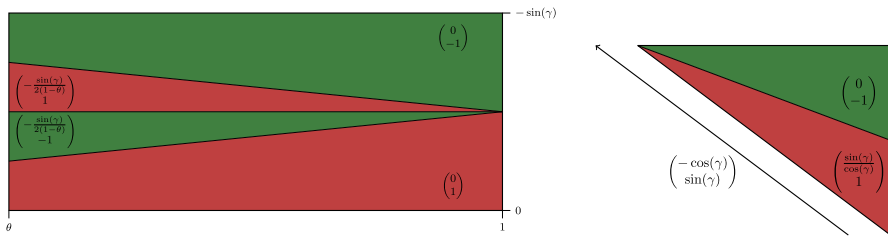


Fig. 5 Left: sketch of the building block for the branching construction on S_y for $N = 2$. Right: sketch of the gradient of the function $u\partial$ used to moderate between the sawtooth functions (in vertical direction) achieved by the branching construction and the boundary condition $u = 0$ on the line $\text{conv}\{(-\cos(\gamma), \sin(\gamma)), (0, 0)\}$. This construction is used if $0 \leq y \leq n/4$ is relatively small. The red and green regions indicate that the y -derivative is ± 1 , respectively (Color figure online)

Then define

$$u_K(x, y) = \frac{1}{M} 2^{-k} u_{bb} \left(\theta^{-k} \left(x + \frac{m}{M} \cos(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right), 2^k M y \right).$$

3. Let $0 \leq k \leq K$, $0 \leq m \leq M - 1$, $a \in \{(a_1, \dots, a_k) : a_i \in \{0, 1\}, 1 \leq i \leq k\}$ and

$$\begin{aligned} (x, y) \in & \left(\theta^{k+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right. \\ & \left. - 2^{-k-1} \frac{\cos(\gamma)}{M}, \theta^{k+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \\ & \times \left(\frac{m}{M} \sin(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-k-1} \frac{\sin(\gamma)}{M}, \frac{m}{M} \sin(\gamma) \right. \\ & \left. + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-k} \frac{\sin(\gamma)}{M} \right). \end{aligned}$$

Then define u_K constant in x -direction as

$$u_K(x, y) = \frac{1}{M} 2^{-k} u_{bb}(\theta, 2^k M y).$$

4. Let $a \in \{(a_1, \dots, a_k) : a_i \in \{0, 1\}, 1 \leq i \leq K + 1\}$, $0 \leq m \leq M - 1$ and

$$(x, y) \in \left(-\frac{m}{M} \cos(\gamma) - \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M}, \theta^{K+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \\ \times \left(\frac{m}{M} \sin(\gamma) + \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M}, \frac{m}{M} \sin(\gamma) \right. \\ \left. + \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-K-1} \frac{\sin(\gamma)}{M} \right).$$

Then define u_K constant in x -direction as

$$u_K(x, y) = \frac{1}{M} 2^{-K} u_{bb}(\theta, 2^K M y).$$

5. Let $\ell \in \{0, \dots, 2^{K+1}M - 1\}$ and

$$(x, y) \in \text{conv} \left\{ \frac{\ell}{2^{K+1}M} \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \frac{\ell+1}{2^{K+1}M} \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \frac{\ell}{2^{K+1}M} \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix} \right. \\ \left. + \frac{1}{2^{K+1}M} \begin{pmatrix} 0 \\ \sin(\gamma) \end{pmatrix} \right\}.$$

Then we define

$$u_K(x, y) = \frac{2^{-K-1}}{M} u_{\partial} \left(2^{K+1}M \left(x + \frac{\ell}{2^{K+1}M} \cos(\gamma) \right), 2^{K+1}M \left(y - \frac{\ell}{2^{K+1}M} \sin(\gamma) \right) \right).$$

We note that by definition it holds that $u_K(\cdot, 0) = u_K(\cdot, \sin(\gamma)) = 0$ and $u_K(-t \cos(\gamma), t \sin(\gamma)) = 0$ for all $t \in (0, 1)$. Moreover, we estimate using that $\cos(\gamma) \geq \cos(\pi/4) > 0$, $\partial_1 u_K = 0$ a.e. in regions (1), (3) and (4), and $\partial_2 u_K \in \{\pm 1\}$ a.e.

$$\int_{S_{\gamma}} (\partial_1 u_K)^2 + \min\{|\partial_2 u_K - 1|, |\partial_2 u_K + 1|\}^2 dx dy \\ \leq \sum_{k=0}^K \frac{1}{M^2} (4\theta)^{-k} \int_{\theta}^1 \int_0^{\sin(\gamma)} (\partial_1 u_{bb}(x, y))^2 dx dy + \frac{1}{M} 2^{-K-1} \int_{\Delta_{\gamma}} (\partial_1 u_{\partial}(x, y))^2 dx dy \\ \leq C \left(\frac{1}{M^2} \sin(\gamma)^3 + \frac{1}{M} 2^{-K-1} \sin(\gamma)^3 \right) \quad (4)$$

and

$$|D^2 u_K| (S_{\gamma}) \leq |\partial_2 \partial_2 u_K| (S_{\gamma}) + 2 |\partial_2 \partial_1 u_K| (S_{\gamma}) + |\partial_1 \partial_1 u_K| (S_{\gamma}) \\ \leq C \left(\sum_{k=0}^{K+1} M 2^k \theta^k + K + 1 + \sin(\gamma) K + \sum_{k=0}^{K+1} 2^{-k} \theta^{-k} \sin(\gamma) \right)$$

$$\begin{aligned}
&\leq C \left(M + K + 2^{-K} \theta^{-K} \sin(\gamma) \right) \\
&\leq C \left(M + K + 2^K \sin(\gamma) \right).
\end{aligned} \tag{5}$$

Eventually, choose $M = \lceil \sigma^{-1/3} \sin(\gamma) \rceil \leq 2\sigma^{-1/3} \sin(\gamma)$ and K such that $K = \lceil \frac{\log(\sigma^{-1/3} \sin(\gamma))}{\log(2)} \rceil \leq 2 \frac{\sigma^{-1/3} \sin(\gamma)}{\log(2)}$ (recall that $\sigma \leq \sin(\gamma)^3$). It then follows from (4) and (5)

$$\begin{aligned}
&E_{\sigma, \pi/2, 2}(u_K; S_\gamma) \\
&= \int_{S_\gamma} (\partial_1 u_K)^2 + \min\{|\partial_2 u_K - 1|, |\partial_2 u_K + 1|\}^2 dx dy + \sigma |D^2 u_K| (S_\gamma) \\
&\leq C \left(\sigma^{2/3} \sin(\gamma) + \sigma^{1/3} 2^{-K} \sin(\gamma)^2 + \sigma K + \sigma 2^K \sin(\gamma) \right) \\
&\leq C \sigma^{2/3} \sin(\gamma).
\end{aligned}$$

Step 1b: Branching on S_γ for $\gamma \in (\pi/4, \pi/2]$:

Next, consider $\frac{\pi}{4} < \gamma \leq \frac{\pi}{2}$. Again, fix $K, M \in \mathbb{N}$ whose role is the same as above. In the regions (1), (2), (3), we define the function $u_K : S_\gamma \rightarrow \mathbb{R}$ as before. Precisely (see Fig. 4),

1. Let $0 \leq m \leq M - 1$. Then define for $(x, y) \in (1 - \frac{m}{M} \cos(\gamma), 1) \times (\frac{m \sin(\gamma)}{M}, \frac{(m+1) \sin(\gamma)}{M})$

$$u_K(x, y) = \frac{1}{M} u_{bb}(1, My).$$

2. Let $0 \leq k \leq K, 0 \leq m \leq M - 1, a \in \{(a_1, \dots, a_k) : a_i \in \{0, 1\}, 1 \leq i \leq k\}$ and

$$\begin{aligned}
&(x, y) \in \left(\theta^{k+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M}, \theta^k - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \\
&\quad \times \left(\frac{m}{M} \sin(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M}, \frac{m}{M} \sin(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-k} \frac{\sin(\gamma)}{M} \right).
\end{aligned}$$

Then define

$$u_K(x, y) = \frac{1}{M} 2^{-k} u_{bb} \left(\theta^{-k} \left(x + \frac{m}{M} \cos(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right), 2^k My \right).$$

3. Let $0 \leq k \leq K, 0 \leq m \leq M - 1, a \in \{(a_1, \dots, a_k) : a_i \in \{0, 1\}, 1 \leq i \leq k\}$ and

$$\begin{aligned}
(x, y) \in & \left(\theta^{k+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right. \\
& \left. - 2^{-k-1} \frac{\cos(\gamma)}{M}, \theta^{k+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \\
& \times \left(\frac{m}{M} \sin(\gamma) + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-k-1} \frac{\sin(\gamma)}{M}, \frac{m}{M} \sin(\gamma) \right. \\
& \left. + \sum_{k'=1}^k a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-k} \frac{\sin(\gamma)}{M} \right).
\end{aligned}$$

Then define

$$u_K(x, y) = \frac{1}{M} 2^{-k} u_{bb}(\theta, 2^k M y).$$

However, in this case we simply use linear interpolation in x to a vertical interface in region (4) and then extend u_K by 0 in region (5). Precisely:

(4) Let $a \in \{(a_1, \dots, a_k) : a_i \in \{0, 1\}, 1 \leq i \leq K+1\}$, $0 \leq m \leq M-1$ and

$$\begin{aligned}
(x, y) \in & \left(-\frac{m}{M} \cos(\gamma) - \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M}, \theta^{K+1} - \frac{m}{M} \cos(\gamma) - \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \\
& \times \left(\frac{m}{M} \sin(\gamma) + \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M}, \frac{m}{M} \sin(\gamma) + \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\sin(\gamma)}{M} + 2^{-K-1} \frac{\sin(\gamma)}{M} \right).
\end{aligned}$$

Then define

$$u_K(x, y) = \theta^{-K-1} \left(x + \frac{m}{M} \cos(\gamma) + \sum_{k'=1}^{K+1} a_{k'} 2^{-k'} \frac{\cos(\gamma)}{M} \right) \frac{1}{M} 2^{-K} u_{bb}(\theta, 2^K M y).$$

(5) Let $\ell \in \{0, \dots, 2^{K+1}M-1\}$ and

$$\begin{aligned}
(x, y) \in & \text{conv} \left\{ \frac{\ell}{2^{K+1}M} \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \frac{\ell+1}{2^{K+1}M} \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \frac{\ell}{2^{K+1}M} \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix} \right. \\
& \left. + \frac{1}{2^{K+1}M} \begin{pmatrix} 0 \\ \sin(\gamma) \end{pmatrix} \right\}.
\end{aligned}$$

Then we define

$$u_K(x, y) = 0.$$

We only compute the energy contributions from the regions (4) and (5). The contribution in (4) for the term $\sigma |D^2 u_K|$ can be estimated by a term of the form $C\sigma(M2^K\theta^K + 1)$, whereas the other term can be estimated by $C((4\theta)^{-K}M^{-2}\sin(\gamma)^3 + \theta^K\sin(\gamma))$ since the building block u_{bb} is bounded by $\sin(\gamma)$. Additionally, in region (5) we obtain an energy less than $2^{-K}\frac{1}{M}$. Hence, we find

$$\begin{aligned} & \int_{S_\gamma} (\partial_1 u_K)^2 + \min\{|\partial_2 u_K - 1|, |\partial_2 u_K + 1|\}^2 dx dy + \sigma |D^2 u_K| (S_\gamma) \\ & \leq C \left(\frac{1}{M^2} + 2^{-K} \frac{1}{M} + \theta^K \right) + C\sigma(M + K + 2^K) \\ & \leq C \left(\frac{1}{M^2} + 2^{-K} \frac{1}{M} + \sigma(M + 2^K) \right). \end{aligned}$$

Then choose $M = \lceil \sigma^{-1/3} \rceil$ and $K = \frac{\log \sigma^{-1/3}}{\log(2)} \leq 2 \frac{\sigma^{-1/3}}{\log(2)}$. It follows that

$$\begin{aligned} & E_{\sigma, \pi/2, 2}(u_K; S_\gamma) \\ & = \int_{S_\gamma} (\partial_1 u_K)^2 + \min\{|\partial_2 u_K - 1|, |\partial_2 u_K + 1|\}^2 dx dy + \sigma |D^2 u_K| (S_\gamma) \\ & \leq C\sigma^{2/3} \leq C\sigma^{2/3} \sin(\gamma), \end{aligned}$$

since $\pi/4 \leq \gamma \leq \pi/2$.

Step 1c: Branching on $(0, 1)^2$:

Define the set

$$T_\gamma = \left\{ (x, y) \in (0, 1)^2 : (x, y - 1) \cdot e^{i\gamma} \leq 0 \right\}.$$

Let $w : S_\gamma \rightarrow \mathbb{R}$ be the function from step 1a or 1b according to the case $\gamma \in (\pi/4, \pi/2]$ or $\gamma \in (0, \pi/4]$, respectively. Note that $w(x, \sin(\gamma)) = 0$ for all $x \in (-\cos(\gamma), 1)$. Then we define (see Fig. 3)

$$u(x, y) = \begin{cases} e^{i\gamma} \cdot (x, y - 1) & \text{if } (x, y) \notin T_\gamma, \\ w(R_\gamma(x, y - 1)) & \text{if } (x, y) \in T_\gamma, \end{cases}$$

where R_γ is the rotation by the angle $\pi/2 - \gamma$. Using step 1, we obtain that

$$E_{\sigma, \gamma, 2}(u) \leq E_{\sigma, \pi/2, 2}(w; S_\gamma) + 4\sigma \leq C |\sin(\gamma)| \sigma^{2/3} + 4\sigma \leq C |\sin(\gamma)| \sigma^{2/3},$$

where we used that $\sigma^{1/3} \leq |\sin(\gamma)|$.

Step 2: Lower bound:

Let $u \in \mathcal{A}$ and assume for simplicity $-\pi/2 \leq \gamma \leq 0$. Again, note that for $\sigma \geq |\sin(\gamma)|^3$ the lower bound is shown in Proposition 4, c.f. Remark 1. Hence, we assume

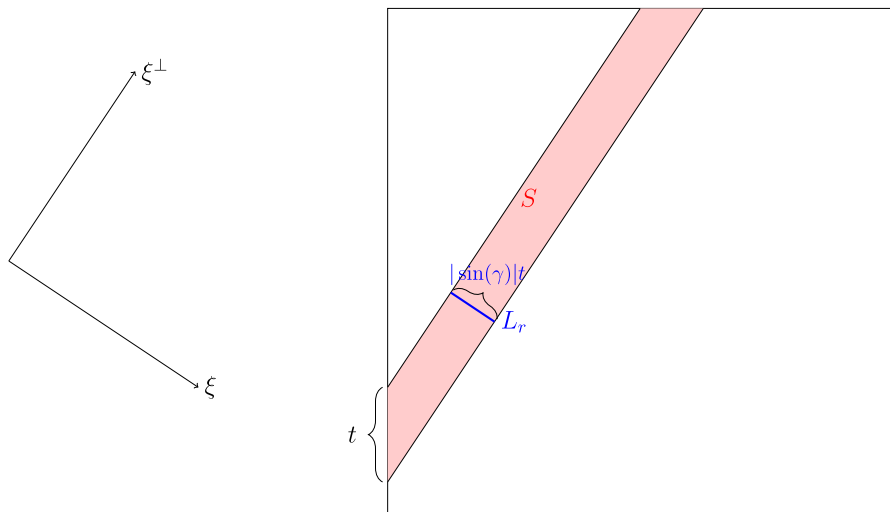


Fig. 6 Sketch of the choices made in the proof of the lower bound in Theorem 1 for $N = 2$

that $\sigma \leq |\sin(\gamma)|^3$. Now, let $t := \frac{1}{4}\sigma^{1/3} |\sin(\gamma)|^{-1} \leq 1/4$, $\xi := e^{i\gamma}$ and $\xi^\perp := e^{i\gamma+i\pi/2}$. Then find $a \in (t, \frac{1}{2})$ such that (see Fig. 6)

$$E_{\sigma,\gamma,2} \left(u; (0, 1)^2 \cap \{(0, y) + r\xi^\perp : y \in (a - t, a), r \in (0, 1)\} \right) \leq 4t E_{\sigma,\gamma,2}(u).$$

Let us write $S := (0, 1)^2 \cap \{(0, y) + r\xi^\perp : y \in (a - t, a), r \in (0, 1)\}$. Now, we write for $r \in (0, 1/2)$

$$L_r = \left\{ a + r\xi^\perp + s\xi : s \in (0, |\sin(\gamma)|t) \right\} \subseteq S.$$

Then find by a slicing and Fubini-type argument $r \in (0, 1/2)$ such that

$$\int_{L_r} \text{dist}(\nabla u, K_{\gamma,N})^2 d\mathcal{H}^1 + \sigma |\partial_\xi \nabla u| (L_r) \leq 2E_{\sigma,\gamma,2}(u; S) \leq (8t)E_{\sigma,\gamma,2}(u). \quad (6)$$

Now one of the following options holds true:

1. $\text{dist}(\nabla u, K_{\gamma,N}) = |\nabla u - \xi|$ almost everywhere on L_r ,
2. $\text{dist}(\nabla u, K_{\gamma,N}) = |\nabla u + \xi|$ almost everywhere on L_r ,
3. $|\partial_\xi \nabla u| (L_r) \geq 1/4$,
4. $\text{dist}(\nabla u, K_{\gamma,N}) \geq 1/4$ almost everywhere on L_r .

If 3. or 4. holds true, we obtain from (6)

$$\min\{\sigma/4, |\sin(\gamma)|t/16\} \leq 8tE_{\sigma,\gamma,2}(u). \quad (7)$$

Let us now assume that 1. holds true (case 2. can be treated similarly). By the triangle inequality, we obtain

$$\frac{1}{4}(|\sin(\gamma)|t)^2 = \frac{1}{4}\mathcal{H}^1(L_r)^2 \leq \min_{c \in \mathbb{R}} \|u - \xi \cdot (x, y) - c\|_{L^1(L_r)} + \|u\|_{L^1(L_r)}.$$

On the other hand, we estimate using Poincaré's inequality

$$\begin{aligned} \min_{c \in \mathbb{R}} \|u(x, y) - \xi \cdot (x, y) - c\|_{L^1(L_r)} &\leq (|\sin(\gamma)|t) \|\partial_\xi u(x, y) - 1\|_{L^1(L_r)} \\ &\leq (|\sin(\gamma)|t)^{3/2} \|\nabla u(x, y) - \xi\|_{L^2(L_r)} \\ &\leq \sqrt{8}(|\sin(\gamma)|t)^{3/2} t^{1/2} E_{\sigma, \gamma, 2}(u)^{1/2} \end{aligned}$$

and

$$\begin{aligned} \|u\|_{L^1(L_r)} &\leq \|\partial_\xi^\perp u\|_{L^1(S)} \leq (|\sin(\gamma)|t)^{1/2} \|\text{dist}(\nabla u, K_{\gamma, N})\|_{L^2(S)} \\ &\leq 2(|\sin(\gamma)|t)^{1/2} t^{1/2} E_{\sigma, \gamma, 2}(u)^{1/2}. \end{aligned}$$

Consequently, we obtain

$$E_{\sigma, \gamma, 2}(u) \geq \frac{1}{128} \min\{|\sin(\gamma)|, |\sin(\gamma)|^3 t^2\}.$$

Together with (7) this yields

$$E_{\sigma, \gamma, N}(u) \geq \frac{1}{128} \min\{\sigma t^{-1}, |\sin(\gamma)|, |\sin(\gamma)|^3 t^2\} \geq \frac{1}{2048} \sigma^{2/3} |\sin(\gamma)|,$$

since $|\sin(\gamma)| \geq |\sin(\gamma)|^3 \geq |\sin(\gamma)| \sigma^{2/3}$. $\square$

6 The Problem with Three or Four Gradients

6.1 The Lower Bound

We split the proof of the lower bound into different steps depending on the size of $|\sin(\gamma)|$. This general strategy is similar to the one used in Ginster and Zwicknagl (2023), c.f. also Ginster and Zwicknagl (2024). However, in contrast with Ginster and Zwicknagl (2023) in the setting of this manuscript the geometry of the set $K_{\gamma, N}$ has to be reflected in the lower bound argument. In addition, the arguments in Ginster and Zwicknagl (2023) do not lead to the length scales which appear naturally in the upper bound constructions. In this manuscript, the technical details of the used arguments are refined which leads to a simpler presentation and the appearance of the length scales from the upper bound constructions also in the proof of the lower bound.

We start with proving the lower bound for all relatively large (in absolute value) angles $\gamma \in \Gamma_N$. Here, the term $|\log |\sin(\gamma)||$ in the denominator of the lower bound will not play a role as it is uniformly bounded.

Proposition 5 Let $N = 3, 4$ and $\gamma_0 \in \Gamma_N$ with $\gamma_0 > 0$. Then there exists a constant $c_B > 0$ such that for all $\gamma \in \Gamma_N$ with $|\gamma| \geq \gamma_0$ it holds

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \geq c_B \min\{\sigma(|\log \sigma| + 1), 1\}.$$

Proof Let $u \in \mathcal{A}$. We fix $\alpha_0 = \min\{\sin(\gamma_0), \cos(\pi/4), 1/16\}$ and $c_1 = \frac{\alpha_0^2}{256} \leq 1$. We assume that $E_{\sigma, \gamma, N}(u) \leq \frac{c_1^{1/2}}{16} \min\{\sigma(|\log \sigma| + 1), 1\}$. Otherwise there is nothing to show.

Let $0 < s \leq c_1$. Then find an interval $I \subseteq (0, 1)$ of length $\frac{s}{c_1}$ such that

$$\begin{aligned} E_{\sigma, \gamma, N}(u; (0, 1) \times I) &\leq \frac{8s}{c_1} E_{\sigma, \gamma, N}(u) \text{ and} \\ &\int_I \text{dist}(\nabla u(s, t), K_{\gamma, N})^2 dt + \sigma |\partial_2 \nabla u(s, \cdot)| (I) \\ &\leq \frac{8s}{c_1} \left(\int_0^1 \text{dist}(\nabla u(s, t), K_{\gamma, N})^2 dt + \sigma |\partial_2 \nabla u(s, \cdot)| (0, 1) \right). \end{aligned}$$

Then one of the following three conditions has to hold on I :

1. $|\partial_2 \nabla u(s, \cdot)| (I) \geq \frac{1}{4}$,
2. $\exists \xi \in K_{\gamma, N}$ such that $|\nabla u(s, t) - \xi| \leq \frac{1}{2}$ for almost all $t \in I$,
3. $\int_I \text{dist}(\nabla u(s, t), K_{\gamma, N})^2 dt \geq \frac{1}{16} |I|$.

If 1. or 3. is true, we obtain

$$\int_I \text{dist}(\nabla u(s, t), K_{\gamma, N})^2 dt + \sigma |\partial_2 \nabla u(s, \cdot)| (I) \geq \min\{\sigma/4, |I|/16\}.$$

Lastly, we assume that 2. is true. By the triangle inequality, we observe (recall that $|\xi_2| \geq \sin(\gamma_0)$ by definition of $K_{\gamma, N}$)

$$\frac{\alpha_0}{4} |I|^2 \leq \min_{c \in \mathbb{R}} \|\xi_2 y - c\|_{L^1(I)} \leq \|u(s, \cdot)\|_{L^1(I)} + \min_{c \in \mathbb{R}} \|u(s, y) - \xi_2 y - c\|_{L^1(I)}.$$

Let us first assume that $\frac{\alpha_0}{8} |I|^2 \leq \|u(s, \cdot)\|_{L^1(I)}$. We estimate

$$\begin{aligned} \frac{\alpha_0}{8} |I|^2 &\leq \int_I |u(s, t)| dt \leq \int_0^s \int_I \text{dist}(\nabla u(s, t), K_{\gamma, N}) dt ds + s |I| \\ &\leq E_{\sigma, \gamma, N}(u; (0, 1) \times I)^{1/2} s^{1/2} |I|^{1/2} + s |I|. \end{aligned}$$

By the definition of c_1 it follows $s \leq \frac{\alpha_0}{16} |I|$. Consequently, we obtain

$$\frac{\alpha_0}{16} |I|^2 \leq E_{\sigma, \gamma, N}(u)^{1/2} (8 |I|)^{1/2} \left(\frac{\alpha_0}{16} \right)^{1/2} |I|,$$

which in turn yields

$$\frac{1}{8c_1^{1/2}}s \leq \frac{\alpha_0}{128} |I| \leq E_{\sigma,\gamma,N}(u) \leq \frac{c_1^{1/2}}{16} \min\{1, \sigma(|\log \sigma| + 1)\},$$

i.e., $s \leq \frac{c_1}{2} \min\{1, \sigma(|\log \sigma| + 1)\}$.

On the other hand, if $\frac{\alpha_0}{8} |I|^2 \leq \min_{c \in \mathbb{R}} \|u(s, y) - \xi_2 y - c\|_{L^1(I)}$, we estimate using Poincaré's and Hölder's inequality

$$\begin{aligned} \frac{\alpha_0}{8} |I|^2 &\leq \min_{c \in \mathbb{R}} \|u(s, y) - \xi_2 y - c\|_{L^1(I)} \\ &\leq |I| \|\partial_2 u(s, y) - \xi_2\|_{L^1(I)} \\ &\leq |I|^{3/2} \left(\int_I \text{dist}(\nabla u(s, y), K_{\gamma,N})^2 dy \right)^{1/2}. \end{aligned}$$

Here, we used that 2. implies that $|\nabla u(s, t) - \xi| = \text{dist}(\nabla u(s, t), K_{\gamma,N})$ for a.e. $t \in I$. It follows that

$$\frac{\alpha_0^2}{64} |I| \leq \int_I \text{dist}(\nabla u(s, y), K_{\gamma,N})^2 dy.$$

Combining the cases 1., 2. and 3. we obtain for all $\frac{c_1}{2} \min\{1, \sigma(|\log \sigma| + 1)\} \leq s \leq c_1$

$$\begin{aligned} 8 |I| &\left(\int_0^1 \text{dist}(\nabla u(s, y), K_{\gamma,N})^2 dy + \sigma |\partial_2 \nabla u(s, \cdot)| (0, 1) \right) \\ &\geq \int_I \text{dist}(\nabla u(s, y), K_{\gamma,N})^2 dy + \sigma |\partial_2 \nabla u(s, \cdot)| (I) \\ &\geq \min \left\{ \frac{\sigma}{4}, \frac{\alpha_0^2}{64} |I| \right\}. \end{aligned}$$

Consequently, we find

$$\begin{aligned} E_{\sigma,\gamma,N}(u) &\geq \int_{\frac{c_1}{2} \min\{1, \sigma(|\log \sigma| + 1)\}}^{c_1} \left(\int_0^1 \text{dist}(\nabla u(s, y), K_{\gamma,N})^2 dy + \sigma |\partial_2 \nabla u(s, \cdot)| (0, 1) \right) ds \\ &\geq \int_{\frac{c_1}{2} \min\{1, \sigma(|\log \sigma| + 1)\}}^{c_1} \min \left\{ \frac{c_1 \sigma}{32s}, \frac{\alpha_0^2}{512} \right\} ds. \end{aligned} \quad (8)$$

Let us first assume $\sigma \leq \sigma_0$ for some fixed $0 < \sigma_0 \leq 1$. In addition, note that $\frac{\alpha_0^2}{512} \geq \frac{\alpha_0^2 c_1 \sigma}{32 \cdot 32s}$ for all $s \geq \frac{c_1}{2} \sigma$. Consequently, (8) implies that

$$E_{\sigma,\gamma,N}(u) \geq \frac{\alpha_0^2}{32} \int_{\frac{c_1}{2} \sigma (|\log \sigma| + 1)}^{c_1} \frac{c_1 \sigma}{32s} ds$$

$$\begin{aligned}
&= \frac{c_1 \alpha_0^2}{1024} \sigma (|\log \sigma| + \log(2) - \log(|\log \sigma| + 1)) \\
&\geq \frac{c_1 \alpha_0^2 \log(2)}{2048} \sigma (|\log \sigma| + 1)
\end{aligned}$$

if σ_0 is small enough.

On the other hand, assume $\sigma > \sigma_0$. Then it follows from (8)

$$E_{\sigma, \gamma, N}(u) \geq \int_{c_1/2}^{c_1} \min \left\{ \frac{\sigma_0}{32}, \frac{\alpha_0^2}{512} \right\} dt = \frac{c_1}{2} \min \left\{ \frac{\sigma_0}{32}, \frac{\alpha_0^2}{512} \right\}.$$

Hence, setting $c_B = \min \left\{ \frac{c_1 \sigma_0}{64}, \frac{c_1 \alpha_0^2 \log(2)}{2048} \right\}$ the assertion follows. $\square$

Next, we prove the lower bound for all small (in absolute value) angles and small σ . The strategy of the proof is inspired by the proof of Ginster and Zwicknagl (2023, Lemma 6), see also Ginster and Zwicknagl (2024). However, the details presented here are refined and lead to a simpler presentation.

Proposition 6 *Let $N = 3, 4$. There exists $\pi/4 > \gamma_0 > 0$, $K \in \mathbb{N}$ and a constant $c_D > 0$ such that for all $\gamma \in \Gamma_N$ with $|\gamma| \leq \gamma_0$ and $\sigma \leq |\sin(\gamma)|^K$ it holds*

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \geq c_D \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right).$$

Proof *Step 1: Preliminaries.* Fix $K = 32$. Moreover, let $\gamma_0 > 0$ be such that $k \sin(\gamma_0)^{k/4} \leq \frac{1}{3 \cdot 256 \cdot 9 \cdot 48 \cdot 120}$ for all $k \geq K$. Let $u \in \mathcal{A}$. Let $k > K$ and assume that $\sigma \in (|\sin(\gamma)|^k, |\sin(\gamma)|^{k-1})$. Then $\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \leq k + 1 \leq 2k$. Hence, it suffices to prove that $E_{\sigma, \gamma, N}(u) \geq ck\sigma$ for a universal constant $c > 0$. In particular, we may assume that $E_{\sigma, \gamma, N}(u) \leq k\sigma$.

Next, note that by the choice of γ_0 it follows for all $|\gamma| \leq \gamma_0$ that $|\sin(\gamma)| \leq \frac{1}{4}$ which implies $\frac{1}{2} |\sin(\gamma)|^i - \frac{3}{2} |\sin(\gamma)|^{i+1} \geq \frac{1}{8} |\sin(\gamma)|^i$. In particular, the intervals $(\frac{1}{2} |\sin(\gamma)|^i, \frac{3}{2} |\sin(\gamma)|^i)$ are pairwise disjoint. Then find for $i \leq \lfloor k/4 \rfloor$ a point $x_i \in (\frac{1}{2} |\sin(\gamma)|^i, \frac{3}{2} |\sin(\gamma)|^i)$ such that $\nabla u(x_i, \cdot) \in BV((0, 1); \mathbb{R}^2)$ and

$$E_{\sigma, \gamma, N}(u; \{x_i\} \times (0, 1)) \leq |\sin(\gamma)|^{-i} E_{\sigma, \gamma, N} \left(u; \left(\frac{1}{2} |\sin(\gamma)|^i, \frac{3}{2} |\sin(\gamma)|^i \right) \times (0, 1) \right).$$

Claim: For all $i \leq \lfloor k/4 \rfloor$, there exists a constant $c > 0$ such that it holds

$$|\sin(\gamma)|^i E_{\sigma, \gamma, N}(u, \{x_i\} \times (0, 1)) + E_{\sigma, \gamma, N}(u; (x_{i+1}, x_i) \times (0, 1)) \geq c\sigma.$$

Before we prove the claim, we briefly show that it implies the desired lower bound. Indeed, we have

$$\begin{aligned} E_{\sigma,\gamma,N}(u) &\geq \frac{1}{2} \sum_{i=1}^{\lfloor k/4 \rfloor} E_{\sigma,\gamma,N} \left(u, \left(\frac{1}{2} |\sin(\gamma)|^i, \frac{3}{2} |\sin(\gamma)|^i \right) \times (0, 1) \right) \\ &\quad + E_{\sigma,\gamma,N}(u; (x_{i+1}, x_i) \times (0, 1)) \\ &\geq \frac{1}{2} \sum_{i=1}^{\lfloor k/4 \rfloor} |\sin(\gamma)|^i E_{\sigma,\gamma,N}(u, \{x_i\} \times (0, 1)) \\ &\quad + E_{\sigma,\gamma,N}(u; (x_{i+1}, x_i) \times (0, 1)) \geq \frac{c}{16} k\sigma. \end{aligned}$$

Hence, it only remains to show the claim.

Step 2: Estimates on vertical slices. Let $t = 120 |\sin(\gamma)|^i$. Then find $y \in (0, 1)$ with $(y, y+t) \subseteq (0, 1)$ such that

$$\begin{aligned} E_{\sigma,\gamma,N}(u; (0, 1) \times (y, y+t)) &\leq 48t E_{\sigma,\gamma,N}(u), \\ E_{\sigma,\gamma,N}(u; (x_{i+1}, x_i) \times (y, y+t)) &\leq 48t E_{\sigma,\gamma,N}(u; (x_{i+1}, x_i) \times (0, 1)) \\ E_{\sigma,\gamma,N}(u; \{x_i\} \times (y, y+t)) &\leq 48t E_{\sigma,\gamma,N}(u; \{x_i\} \times (0, 1)) \\ \text{and } E_{\sigma,\gamma,N}(u; \{x_{i+1}\} \times (y, y+t)) &\leq 48t E_{\sigma,\gamma,N}(u; \{x_{i+1}\} \times (0, 1)). \end{aligned}$$

Then one of the following holds in $(y, y+t)$ (recall that the minimal distance between two points in $K_{\gamma,N}$ is $\sqrt{2}$)

1. $|\partial_2 \nabla u(x_i, \cdot)| (y, y+t) \geq \frac{1}{4}$,
2. $\exists \xi \in K_{\gamma,N}$ such that $\text{dist}(\nabla u(x_i, \cdot), K_{\gamma,N}) \leq 3 |\nabla u(x_i, \cdot) - \xi|$.

Suppose first that 1. holds. Then

$$\begin{aligned} \frac{\sigma}{4} &\leq E_{\sigma,\gamma,N}(u; \{x_i\} \times (y, y+t)) \leq 48t E_{\sigma,\gamma,N}(u; \{x_i\} \times (0, 1)) \\ &= 48 \cdot 120 |\sin(\gamma)|^i E_{\sigma,\gamma,N}(u; \{x_i\} \times (0, 1)), \end{aligned}$$

and thus, the claim follows in this case.

For the rest of the proof, we will now assume that 2. holds. First, recall that by the definition of Γ_N we have $|\xi_2| \geq |\sin(\gamma)|$. By the triangle inequality, we obtain

$$\begin{aligned} \frac{1}{4} |\xi_2| t^2 &\leq \|u(x_i, s+t/2) - u(x_i, s) - \xi_2 t/2\|_{L^1(y, y+t/2)} \\ &\quad + \|u(x_i, s+t/2) - u(x_i, s)\|_{L^1(y, y+t/2)}. \end{aligned}$$

Hence, it holds one of the following:

- (a) $\frac{1}{8} |\xi_2| t^2 \leq \|u(x_i, s+t/2) - u(x_i, s) - \xi_2 t/2\|_{L^1(y, y+t/2)}$,
- (b) $\frac{1}{8} |\xi_2| t^2 \leq \|u(x_i, s+t/2) - u(x_i, s)\|_{L^1(y, y+t/2)}$.

Let us first assume that (a) holds. We estimate

$$\begin{aligned}
 & \left| \int_y^{y+t/2} u(x_i, s+t/2) - u(x_i, s) - \xi_2 t/2 \, ds \right| \\
 & \leq \int_y^{y+t/2} \int_s^{s+t/2} |\partial_2 u(x_i, r) - \xi_2| \, dr \, ds \\
 & \leq t^{1/2} \int_y^{y+t/2} \left(\int_y^{y+t} |\partial_2 u(x_i, r) - \xi_2|^2 \, dr \right)^{1/2} \, ds \\
 & \leq \sqrt{9 \cdot 48} t^2 E_{\sigma, \gamma, N}(u; \{x_i\} \times (0, 1))^{1/2} \\
 & \leq \frac{1}{16} t^2 |\sin(\gamma)| \leq \frac{1}{16} |\xi_2| t^2.
 \end{aligned} \tag{9}$$

For the third estimate, we used that by the choice of x_i and γ_0 it holds $E_{\sigma, \gamma, N}(u; \{x_i\} \times (0, 1)) \leq |\sin(\gamma)|^{-i} k \sigma \leq k |\sin(\gamma)|^{k-i-2} |\sin(\gamma)|^2 \leq \frac{1}{9 \cdot 48 \cdot 256} |\sin(\gamma)|^2$. Next, observe that it holds by 2. and Poincaré's inequality for $a = \frac{2}{t} \int_y^{y+t/2} u(x_i, s+t/2) - u(x_i, s) - \xi_2 t/2 \, ds$

$$\begin{aligned}
 & \|u(x_i, s+t/2) - u(x_i, s) - \xi_2 t/2 - a\|_{L^1(y, y+t/2)} \\
 & \leq t \|\partial_2 u(x_i, s+t/2) - \partial_2 u(x_i, s)\|_{L^1(y, y+t/2)} \\
 & \leq t (\|\partial_2 u(x_i, s+t/2) - \xi_2\|_{L^1(y, y+t/2)} + \|\xi_2 - \partial_2 u(x_i, s)\|_{L^1(y, y+t/2)}) \\
 & = t \|\partial_2 u(x_i, s) - \xi_2\|_{L^1(y, y+t)} \\
 & \leq 3t^{3/2} E_{\sigma, \gamma, N}(u; \{x_i\} \times (y, y+t))^{1/2} \\
 & \leq \sqrt{9 \cdot 48} t^2 E_{\sigma, \gamma, N}(u; \{x_i\} \times (0, 1))^{1/2}.
 \end{aligned} \tag{10}$$

Consequently, since $\|u(x_i+t/2, s) - u(x_i, s) - \xi_2 t/2\|_{L^1(y, y+t/2)} \geq \frac{1}{8} |\xi_2| t^2$, it follows from (9) and (10) that

$$\begin{aligned}
 432 \cdot t^4 E_{\sigma, \gamma, N}(u; \{x_i\} \times (0, 1)) & \geq \left(\frac{1}{8} |\xi_2| t^2 - |a| t/2 \right)^2 \geq \frac{t^4}{256} |\xi_2|^2 \\
 & \geq \frac{t^4}{256} |\sin(\gamma)|^2 \\
 & \geq \frac{t^4}{256} |\sin(\gamma)|^{-i} \sigma,
 \end{aligned}$$

which yields the claim if 2. and (a) hold.

Hence, from now on we will assume that (b) holds i.e., $\|u(x_i, s) - u(x_i, s+t/2)\|_{L^1(y, y+t/2)} \geq \frac{1}{8} |\xi_2| t^2$.

Step 3: An estimate for horizontal difference quotients. First, observe that it holds for all $s \in (y, y+t/2)$ using 2.

$$\begin{aligned}
|u(x_i, s + t/2) - u(x_i, s)| &\leq |\xi_2| t/2 + \int_s^{s+t/2} |\partial_2 u(x_i, y) - \xi_2| dy \\
&\leq |\xi_2| t/2 + \left(\frac{t}{2}\right)^{1/2} \left(\int_s^{s+t/2} |\partial_2 u(x_i, y) - \xi_2|^2 dy\right)^{1/2} \\
&\leq |\xi_2| t/2 + 3 \left(\frac{t}{2}\right)^{1/2} \left(\int_s^{s+t/2} \text{dist}(\nabla u(x_i, y), K_{\gamma, N})^2 dy\right)^{1/2} \\
&\leq |\xi_2| t/2 + 15t E_{\sigma, \gamma, N}(u; \{x_i\} \times (0, 1))^{1/2} \\
&\leq |\xi_2| t/2 + 15t \left(|\sin(\gamma)|^{-i} k\sigma\right)^{1/2} \leq |\xi_2| t,
\end{aligned}$$

where we used that $|\sin(\gamma)|^{-i} k\sigma \leq |\sin(\gamma)|^{k-i-1} k \leq \frac{1}{15^{2.4}} |\sin(\gamma)|^2 \leq \frac{1}{15^{2.4}} |\xi_2|^2$. Next, define

$$A_i = \left\{ s \in (y, y+t) : E_{\sigma, \gamma, N}(u; (0, 1) \times \{s\}) \leq \frac{80}{t} E_{\sigma, \gamma, N}(u; (0, 1) \times (y, y+t)) \right\}$$

Then $\mathcal{L}^1(A_i) \geq \frac{79}{80}t$. For $s \in A_i$, we estimate similar to the above using that $K_{\gamma, N} \subseteq S^1$ and $u(0, s) = 0$

$$\begin{aligned}
|u(x_{i+1}, s)| &\leq x_{i+1} + \int_0^{x_{i+1}} \min_{\tilde{\xi} \in K_{\gamma, N}} |\partial_1 u(x, s) - \tilde{\xi}_1| dx \\
&\leq x_{i+1} + x_{i+1}^{1/2} \left(\int_0^{x_{i+1}} \min_{\tilde{\xi} \in K_{\gamma, N}} |\nabla u(x, s) - \tilde{\xi}|^2 dx\right)^{1/2} \\
&\leq x_{i+1} + x_{i+1}^{1/2} E_{\sigma, \gamma, N}(u; (0, 1) \times \{s\})^{1/2} \\
&\leq x_{i+1} + x_{i+1}^{1/2} \left(\frac{80}{t} E_{\sigma, \gamma, N}(u; (0, 1) \times (y, y+t))\right)^{1/2} \\
&\leq x_{i+1} + \sqrt{80 \cdot 48} \cdot x_{i+1}^{1/2} E_{\sigma, \gamma, N}(u)^{1/2} \\
&\leq 2x_{i+1} \\
&\leq \frac{1}{40} |\sin(\gamma)| t \leq \frac{1}{40} |\xi_2| t.
\end{aligned}$$

For the fourth inequality, we used that we have $k\sigma \leq k |\sin(\gamma)|^{k-1-(i+1)} |\sin(\gamma)|^{i+1} \leq \frac{1}{2 \cdot 48 \cdot 80} |\sin(\gamma)|^{i+1} \leq \frac{1}{48 \cdot 80} x_{i+1}$. Next, find for $s \in (y, y+t)$ a value $\bar{s} \in A_i$ such that $|s - \bar{s}| \leq \frac{1}{80}t$. Then we obtain

$$\begin{aligned}
|u(x_{i+1}, s)| &\leq |u(x_{i+1}, s) - u(x_{i+1}, \bar{s})| + |u(x_{i+1}, \bar{s})| \\
&\leq |\xi_2| |s - \bar{s}| + 3 |s - \bar{s}|^{1/2} E_{\sigma, \gamma, N}(u; \{x_{i+1}\} \times (y, y+t))^{1/2} + \frac{1}{40} |\xi_2| t
\end{aligned}$$

$$\leq \frac{1}{20} |\xi_2| t, \quad (11)$$

where we used similarly to above that it holds $E_{\sigma, \gamma, N}(u; \{x_{i+1}\} \times (y, y+t)) \leq \frac{1}{9.80} |\sin(\gamma)|^2 t$. In particular, we obtain for almost all $s \in (y, y+t/2)$ that

$$\begin{aligned} & |u(x_i, s) - u(x_i, s+t/2) - u(x_{i+1}, s) + u(x_{i+1}, s+t/2)| \\ & \leq |u(x_i, s) - u(x_i, s+t/2)| + |u(x_{i+1}, s)| + |u(x_{i+1}, s+t/2)| \\ & \leq 2 |\xi_2| t. \end{aligned} \quad (12)$$

On the other hand, it holds by (b) and (11) that

$$\begin{aligned} & \|u(x_i, s) - u(x_i, s+t/2) - u(x_{i+1}, s) + u(x_{i+1}, s+t/2)\|_{L^1(y, y+t/2)} \\ & \geq \|u(x_i, s) - u(x_i, s+t/2)\|_{L^1(y, y+t/2)} - \|u(x_{i+1}, s)\|_{L^1(y, y+t/2)} \\ & \quad - \|u(x_{i+1}, s+t/2)\|_{L^1(y, y+t/2)} \\ & \geq \frac{1}{8} |\xi_2| t^2 - \frac{1}{20} |\xi_2| t^2 \\ & \geq \frac{1}{16} |\xi_2| t^2. \end{aligned} \quad (13)$$

We define

$$S := \left\{ s \in (y, y+t/2) : \frac{1}{16} |\xi_2| t \leq g_i(s) \leq 2 |\xi_2| t \right\},$$

where $g_i(s) = |u(x_i, s) - u(x_i, s+t/2) - u(x_{i+1}, s) + u(x_{i+1}, s+t/2)|$. Combining (12) and (13) yields

$$\frac{1}{16} |\xi_2| t^2 \leq \left(\frac{t}{2} - \mathcal{L}^1(S) \right) \frac{1}{16} |\xi_2| t + \mathcal{L}^1(S) 2 |\xi_2| t \leq \frac{1}{32} |\xi_2| t^2 + 2 \mathcal{L}^1(S) |\xi_2| t,$$

which implies $\mathcal{L}^1(S) \geq \frac{t}{64}$. Next, note that $x_i - x_{i+1} \leq \frac{3}{2} |\sin(\gamma)|^i$. Hence, for a subset $(y, y+t/2)$ of size at least $\frac{t}{64}$ it holds

$$5 |\xi_2| \leq \left| \frac{u(x_i, s) - u(x_{i+1}, s)}{x_i - x_{i+1}} - \frac{u(x_i, s+t/2) - u(x_{i+1}, s+t/2)}{x_i - x_{i+1}} \right|.$$

It follows for a subset of $(y, y+t)$ whose measure is at least $\frac{t}{64}$ that

$$\left| \frac{u(x_i, s) - u(x_{i+1}, s)}{x_i - x_{i+1}} - \xi_1 \right| \geq \frac{5}{2} |\sin(\gamma)|. \quad (14)$$

Step 4: Lower bound for $E_{\sigma, \gamma, N}(u; (x_{i+1}, x_i) \times (0, 1))$. Let us now fix such an $s \in (y, y+t)$ for which (14) holds. Moreover, let us assume that $|\partial_1 \nabla u(\cdot, s)|$

$(x_{i+1}, x_i) \leq 1/20$. It follows that $|\nabla u(x, s) - \xi| \leq |\nabla u(x_i, s) - \xi| + 1/20$ for a.e. $x \in (x_{i+1}, x_i)$. Let us fix $x \in (x_{i+1}, x_i)$ and $\xi \neq \bar{\xi} \in K_{\gamma, N}$. Note that 2. implies that $|\nabla u(x_i, s) - \bar{\xi}| \geq \frac{1}{4} |\xi - \bar{\xi}|$. Thus, using that $|\xi - \bar{\xi}| \geq 1$ it holds

$$|\nabla u(x, s) - \bar{\xi}| \geq |\nabla u(x_i, s) - \bar{\xi}| - \frac{1}{10} \geq \left(\frac{1}{4} - \frac{1}{20}\right) |\xi - \bar{\xi}| = \frac{1}{5} |\xi - \bar{\xi}|.$$

It follows that

$$|\nabla u(x, s) - \xi| \leq |\nabla u(x, s) - \bar{\xi}| + |\xi - \bar{\xi}| \leq 6 |\nabla u(x, s) - \bar{\xi}|.$$

This shows that $|\nabla u(x, s) - \xi| \leq 6 \text{dist}(\nabla u(x, s), K_{\gamma, N})$ for a.e. $x \in (x_{i+1}, x_i)$. Recalling that $x_i - x_{i+1} \geq \frac{1}{2} |\sin(\gamma)|^i - \frac{3}{2} |\sin(\gamma)|^{i+1} \geq \frac{1}{8} |\sin(\gamma)|^i$ it follows that

$$\begin{aligned} & \int_{x_{i+1}}^{x_i} \text{dist}(\nabla u(x, s), K_{\gamma, N})^2 dx \\ & \geq \frac{1}{36} \int_{x_{i+1}}^{x_i} |\partial_1 u(x, s) - \xi_1|^2 ds \\ & \geq \frac{1}{36(x_i - x_{i+1})} \left(\int_{x_{i+1}}^{x_i} \partial_1 u(x, s) - \xi_1 ds \right)^2 \\ & = \frac{1}{36} (x_i - x_{i+1}) \left(\frac{u(x_i, s) - u(x_{i+1}, s)}{x_i - x_{i+1}} - \xi_1 \right)^2 \\ & \geq \frac{25}{32 \cdot 36} |\sin(\gamma)|^{i+2} \geq \frac{25}{32 \cdot 36} \sigma. \end{aligned}$$

Consequently, we obtain

$$\begin{aligned} & 48t E_{\sigma, \gamma, N}(u; (x_{i+1}, x_i) \times (0, 1)) \\ & \geq \int_y^{y+t} \int_{x_{i+1}}^{x_i} \text{dist}(\nabla u(x, s), K_{\gamma, N})^2 dx + \sigma |\partial_1 \nabla u(\cdot, s)| (x_{i+1}, x_i) \\ & \geq \frac{t}{64} \sigma \min \left\{ \frac{25}{32 \cdot 36}, \frac{1}{20} \right\}, \end{aligned}$$

which shows the claim if 2. and (b) hold. This finishes the proof of the claim. $\square$

Eventually, we can now prove the lower bound stated in Theorem 1 for $N = 3, 4$.

Proposition 7 *Let $N = 3, 4$. Then there exists a constant $c > 0$ such that for all $\gamma \in \Gamma_N$ it holds*

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \geq c \min \left\{ |\sin(\gamma)|^2, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right) \right\}.$$

Proof Let us first assume that $\sigma \geq |\sin(\gamma)|^2$. By Proposition 4, it follows

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \geq c_A \min\{\sigma, |\sin(\gamma)|^2\} = c_A |\sin(\gamma)|^2.$$

Next, assume that $\sigma \leq |\sin(\gamma)|^2$ and let $\gamma_0 > 0$ and $K \geq 2$ be as in Proposition 6. Then by Proposition 5 there exists $c_B > 0$ such that for all $\gamma \in \Gamma_N$ with $|\gamma| \geq \gamma_0$,

$$\begin{aligned} E_{\sigma, \gamma, N}(u) &\geq c_B \min\{|\log \sigma| + 1, 1\} \\ &\geq c_B \min\{|\log |\sin(\gamma_0)||, 1\} \min\left\{\sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1\right), |\sin(\gamma)|^2\right\}. \end{aligned}$$

For $\gamma \in \Gamma_N$ with $|\gamma| \leq \gamma_0$ and $\sigma \leq |\sin(\gamma)|^K$, it holds by Proposition 6

$$E_{\sigma, \gamma, N}(u) \geq c_D \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1\right).$$

Eventually, we notice that for $|\sin(\gamma)|^2 \geq \sigma \geq |\sin(\gamma)|^K$ it holds by Proposition 4

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \geq c_A \min\{\sigma, |\sin(\gamma)|^2\} = \frac{c_A}{K+1} \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1\right).$$

This concludes by Remark 1 the proof of the lower bound for $c = \min\left\{\frac{c_A}{K+1}, c_B \min\{1, |\log |\sin(\gamma_0)||\}, c_D\right\}$. $\square$

6.2 The Upper Bound

In this section, we prove the upper bound stated in Theorem 1 for $N = 3, 4$. By Remark 1, it remains to consider the case $\sigma \leq |\sin(\gamma)|^3$. For heuristics on the construction, we refer to the introduction.

Proposition 8 *Let $N = 3, 4$. Then there exists $C > 0$ such that for all $\sigma > 0$ and $\gamma \in \Gamma_N$ it holds*

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \leq C \min\left\{|\sin(\gamma)|^2, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1\right)\right\}.$$

Proof By Proposition 4, we find for $\sigma \geq |\sin(\gamma)|^2$

$$\min_{u \in \mathcal{A}} E_{\sigma, \gamma, N}(u) \leq C_A |\sin(\gamma)|^2 \leq C_A \min\left\{|\sin(\gamma)|^2, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1\right)\right\}.$$

Hence, it remains to consider the case $|\sin(\gamma)|^2 \geq \sigma$. Additionally, we will only treat the case $\gamma \geq 0$. As for $N = 2$, we define the set

$$S_\gamma = \text{conv} \left\{ \begin{pmatrix} -\cos(\gamma) \\ -\sin(\gamma) \end{pmatrix}, \begin{pmatrix} 1 \\ -\sin(\gamma) \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Again, we will first use a self-similar branching construction to define a function on S_γ . However, in this setting the constructed competitor will have gradients in $K_{\pi/2, N}$ in a large part of the domain (c.f. the branching constructions in Ginster and Zwischagl, 2023). Then—up to a rotation—we will use this function on the set $\{(x, y) \in (0, 1)^2 : (x, y - 1) \cdot e^{i\gamma} \leq 0\}$ and glue it to a function with constant gradient $e^{i\gamma}$ on the rest of $(0, 1)^2$.

Step 1: Branching Construction on S_γ for $N = 4$ and $\gamma \in (0, \pi/4) \subseteq \Gamma_4$.

Step 1a: Building block for S_γ . Let $m = \lceil |\sin(\gamma)|^{-1} \rceil$ and define the set

$$B_\gamma^{(4)} = \bigcup_{k=1}^m \left(\frac{\sin(\gamma)}{m} - \frac{k-1}{m} \cos(\gamma), \sin(\gamma) \right) \times \left(\frac{k-1}{m} \sin(\gamma), \frac{k}{m} \sin(\gamma) \right).$$

Then there exists a function $V^{(4)} : B_\gamma^{(4)} \rightarrow \mathbb{R}^2$ such that (see Fig. 7)

- (a) $V^{(4)}(x, y) \in K_{\pi/2, 4}$ for a.e. $(x, y) \in B_\gamma^{(4)}$,
- (b) $|DV^{(4)}|(B_\gamma^{(4)}) \leq C(m \sin(\gamma) + 1) \leq C$,
- (c) For the second component of $V^{(4)}$ it holds (in the sense of traces) the equality $V_2^{(4)}(\frac{\sin(\gamma)}{m} - \frac{k-1}{m} \cos(\gamma), y) = V_2^{(4)}(\sin(\gamma), my - (k-1)\sin(\gamma))$ for all $y \in (\frac{k-1}{m} \sin(\gamma), \frac{k}{m} \sin(\gamma))$,
- (d) $V^{(4)}$ is curl-free,
- (e) $V_1^{(4)}(x, y) = 0$ for all $(x, y) \in \partial B_\gamma$ such that e_1 is tangent to ∂B_γ at (x, y) ,
- (f) It holds $V_2^{(4)}(\sin(\gamma), \cdot) = -\chi_{(0, \sin(\gamma)/2)} + \chi_{(\sin(\gamma)/2, \sin(\gamma))}$.

Step 1b: Interpolation to boundary conditions. We define the function $V_\partial : \text{conv} \left\{ \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \sin(\gamma) \end{pmatrix} \right\} \rightarrow \mathbb{R}^2$ as (c.f. Fig. 4 on the right)

$$V_\partial(x, y) = \begin{cases} \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \text{if } y \geq \sin(\gamma) - \frac{\sin(\gamma)}{2\cos(\gamma)}(x + \cos(\gamma)) \\ \begin{pmatrix} -\frac{\sin(\gamma)}{\cos(\gamma)} \\ -1 \end{pmatrix} & \text{if } y \leq \sin(\gamma) - \frac{\sin(\gamma)}{2\cos(\gamma)}(x + \cos(\gamma)). \end{cases}$$

Step 1c: Branching construction. For $K \in \mathbb{N}$, we define the following function $V_K : S_\gamma \rightarrow \mathbb{R}^2$ as follows (see Fig. 8)

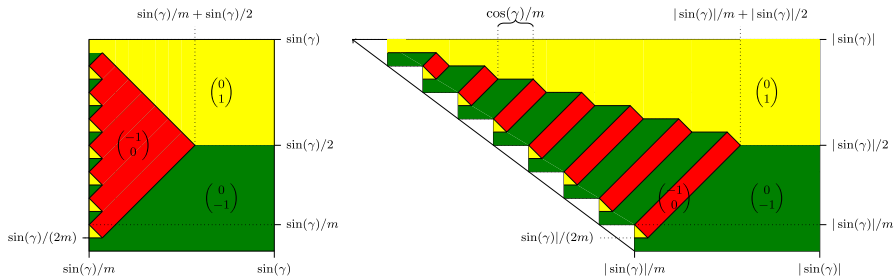


Fig. 7 Sketch for the construction of the building block for $N = 4$ and $m = 8$. Different appearing gradients are color-coded. Left: A version of the construction of the building block for a rectangle of height $\sin(\gamma)$. It is immediate that $|D^2 V^{(3)}| \leq C(\sin(\gamma)m + 1)$. Right: construction of the building block for S_γ , $N = 4$ and $m = 8$. The essential difference to the construction on the left is that after each branching of the construction an extra horizontal gap of length $\cos(\gamma)/m$ has to be bridged by horizontal interfaces creating an extra interface of length $\cos(\gamma)/m$ in horizontal direction and at most $\sqrt{2}\sin(\gamma)$ in diagonal direction. Hence, the total surfaces created in the construction can be estimated up to a constant by $|\sin(\gamma)|m + 1$ (Color figure online)

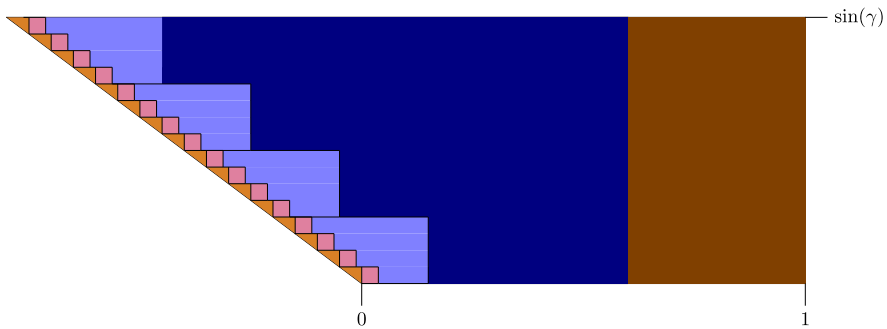


Fig. 8 Sketch of the branching construction for $N \in \{3, 4\}$, $K = 1$ and $m = 4$. Different regions of in the definition of $V_k^{(N)}$ are colored in brown (1), blue (2) (darkblue for $k = 0$, lightblue for $k = 1$), pink (3), and orange (4) (Color figure online)

1. For $x \in (\sin(\gamma), 1)$, we define

$$V_K^{(4)}(x, y) = \begin{cases} \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \text{if } y \geq 1/2, \\ \begin{pmatrix} 0 \\ -1 \end{pmatrix} & \text{if } y \leq 1/2. \end{cases}$$

2. Let $0 \leq k \leq K$ and $l \in \{0, \dots, m^k - 1\}$. Then we define for

$$(x, y) \in \left(-\frac{l}{m^k} \cos(\gamma), \frac{l}{m^k} \sin(\gamma) \right) + \frac{1}{m^k} B_\gamma^{(4)}$$

the function $V_K^{(4)}$ as

$$V_K^{(4)}(x, y) = V^{(4)}\left(m^k(x + \frac{l}{m^k} \cos(\gamma)), m^k(y - \frac{l}{m^k} \sin(\gamma))\right).$$

3. Let $l \in \{0, \dots, m^{K+1} - 1\}$. We define for

$$(x, y) \in \left(-\frac{l}{m^{K+1}} \cos(\gamma), -\frac{l}{m^{K+1}} \cos(\gamma) + \frac{\sin(\gamma)}{m^{K+1}}\right) \times \left(\frac{l}{m^{K+1}} \sin(\gamma), \frac{l+1}{m^{K+1}} \sin(\gamma)\right)$$

the constant extension

$$V_K^{(4)}(x, y) = \begin{cases} \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \text{if } y \geq \frac{l+1/2}{m^{K+1}} \sin(\gamma), \\ \begin{pmatrix} 0 \\ -1 \end{pmatrix} & \text{if } y \leq \frac{l+1/2}{m^{K+1}} \sin(\gamma). \end{cases}$$

4. Let $l \in \{0, \dots, m^{K+1} - 1\}$. We define for

$$(x, y) \in \text{conv} \left\{ \begin{pmatrix} -\frac{l}{m^{K+1}} \cos(\gamma) \\ \frac{l}{m^{K+1}} \sin(\gamma) \end{pmatrix}, \begin{pmatrix} -\frac{l}{m^{K+1}} \cos(\gamma) \\ \frac{l+1}{m^{K+1}} \sin(\gamma) \end{pmatrix}, \begin{pmatrix} -\frac{l+1}{m^{K+1}} \cos(\gamma) \\ \frac{l+1}{m^{K+1}} \sin(\gamma) \end{pmatrix} \right\}$$

the function $V_K^{(4)}$ as

$$V_K^{(4)}(x, y) = V_\partial \left(m^{K+1}(x + \frac{l}{m^{K+1}} \cos(\gamma)), m^{K+1}(y - \frac{l}{m^{K+1}} \sin(\gamma)) \right).$$

We observe that $V_K^{(4)}$ is curl-free as $V^{(4)}$ is curl-free and $\nu \parallel ((V_K^{(4)})^- - (V_K^{(4)})^+)$ on its jump set $J_{V_K^{(4)}}$, where ν is the measure-theoretic normal to $J_{V_K^{(4)}}$, due to (c), (e) and (f). Let $u_K : S_\gamma \rightarrow \mathbb{R}$ be a corresponding primitive with $u_K(0, 0) = 0$. It holds by construction that $u(-t \cos(\gamma), t \sin(\gamma)) = 0$ for all $t \in (0, 1)$ and $u(t, \sin(\gamma)) = 0$ for all $t \in (-\cos(\gamma), 1)$ (c.f. (e)). Next, we estimate using (a), (b) and $\cos(\gamma) \geq \cos(\pi/4) > 0$

$$\begin{aligned} & \int_{S_\gamma} \text{dist}(\nabla u_K, K_{\pi/2,4})^2 \, dx + \sigma \mid D^2 u_K \mid (S_\gamma) \\ & \leq C \frac{\mid \sin(\gamma) \mid^3}{\cos(\gamma) m^{K+1}} + C \sigma (K + \mid \sin(\gamma) \mid + 1) \\ & \leq C \left(\mid \sin(\gamma) \mid^K + \sigma K \right). \end{aligned}$$

Choosing $K = \left\lceil \frac{|\log \sigma|}{|\log \sin(\gamma)|} \right\rceil$ yields the estimate

$$\begin{aligned} E_{\sigma, \pi/2, 4}(u; S_\gamma) &= \int_{S_\gamma} \text{dist}(\nabla u_K, K_{\pi/2, N})^2 dx + \sigma |D^2 u_K| (S_\gamma) \\ &\leq C\sigma \left(\frac{|\log \sigma|}{|\log \sin(\gamma)|} + 1 \right). \end{aligned}$$

Step 2: Branching construction on S_γ for $N = 3$ and $\gamma \in (0, \pi/6) \subseteq \Gamma_3$. The construction for $N = 3$ is very similar to the one for $N = 4$ but needs a different building block.

Step 2a: Building block for S_γ . Let $m = \lceil |\sin(\gamma)|^{-1} \rceil$ and define the set

$$B_\gamma^{(3)} = \bigcup_{k=1}^m \left(\frac{2 \sin(\gamma)}{\sqrt{3}m} - \frac{k-1}{m} \cos(\gamma), \frac{2 \sin(\gamma)}{\sqrt{3}} \right) \times \left(\frac{k-1}{m} \sin(\gamma), \frac{k}{m} \sin(\gamma) \right).$$

Then there exists a function $V^{(3)} : B_\gamma^{(3)} \rightarrow \mathbb{R}^2$ such that (see Fig. 9)

- (a') $V^{(3)}(x, y) \in K_{\pi/2, 3}$ for a.e. $(x, y) \in B_\gamma^{(3)}$,
- (b') $|DV^{(3)}|(B_\gamma^{(3)}) \leq C(m |\sin(\gamma)| + 1) \leq C$,
- (c') For the second component of $V^{(3)}$, it holds $V_2^{(3)}(\frac{2 \sin(\gamma)}{\sqrt{3}m} - \frac{k-1}{m} \cos(\gamma), y) = V_2^{(3)}(\frac{2 \sin(\gamma)}{\sqrt{3}}, my - (k-1) \sin(\gamma))$ for all $y \in (\frac{k-1}{m} \sin(\gamma), \frac{k}{m} \sin(\gamma))$,
- (d') $V^{(3)}$ is curl-free,
- (e') For the first component of $V^{(3)}$ it holds $V_1^{(3)}(x, y) = 0$ for all $(x, y) \in \partial B_\gamma^{(3)}$ such that e_1 is tangent to $\partial B_\gamma^{(3)}$ at (x, y) ,
- (f') It holds $V_2^{(3)}(\frac{2 \sin(\gamma)}{\sqrt{3}}, \cdot) = \chi_{(0, \sin(\gamma)/3)} - \frac{1}{2} \chi_{(\sin(\gamma)/3, \sin(\gamma))}$.

Step 2b: Interpolation to boundary conditions. We define the function $V_\partial^{(3)} : \text{conv} \left\{ \begin{pmatrix} -\cos(\gamma) \\ \sin(\gamma) \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \sin(\gamma) \end{pmatrix} \right\} \rightarrow \mathbb{R}^2$ as

$$V_\partial(x, y) = \begin{cases} \begin{pmatrix} \frac{\sin(\gamma)}{\cos(\gamma)} \\ 1 \end{pmatrix} & \text{if } y \leq \sin(\gamma) - \frac{2 \sin(\gamma)}{3 \cos(\gamma)}(x + \cos(\gamma)) \\ \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} & \text{if } y \geq \sin(\gamma) - \frac{2 \sin(\gamma)}{3 \cos(\gamma)}(x + \cos(\gamma)). \end{cases}$$

Step 2c: Branching construction. For $K \in \mathbb{N}$, we define the function $V_K^{(3)} : S_\gamma \rightarrow \mathbb{R}^2$ as follows (see Fig. 8)

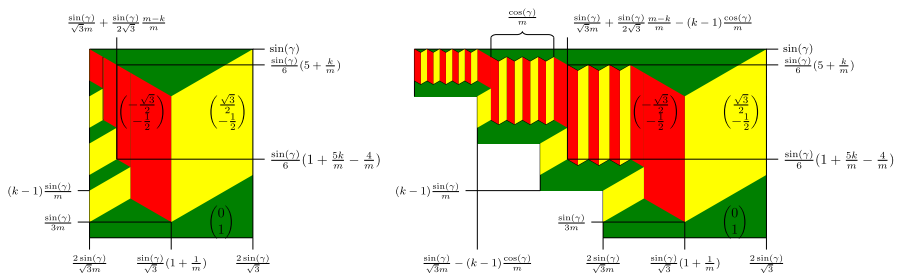


Fig. 9 Sketch for the construction of the building block for Color figure online $N = 3$ and $m = 4$. Different appearing gradients are color-coded, the appearing diagonals have slope $\pm \frac{1}{\sqrt{3}}$. Left: a version of the construction of the building block for a rectangle of height $\sin(\gamma)$. It is immediate that $|D^2 V^{(3)}| \leq C \sin(\gamma)(m+1)$. Right: construction of the building block for $B_\gamma^{(3)}$, $N = 3$ and $m = 4$. The essential difference to the construction on the left is that after each branching of the construction an extra horizontal gap of length $\cos(\gamma)/m$ has to be bridged by sawtooth patterns. In each of those bridging steps, the number of sawteeth can be chosen proportional to the quotient $\frac{\cos(\gamma)}{m} / \left(\frac{\sin(\gamma)}{6} \left(1 + \frac{5k}{m} - \frac{4}{m} \right) - (k-1) \frac{\sin(\gamma)}{m} \right) \leq C \frac{m}{(m-k+2) \sin(\gamma)}$ (in this way $V^{(3)}$ stays $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ on the horizontal parts of the boundary of $B_\gamma^{(3)}$). The corresponding height of each of those vertical interfaces is $\sin(\gamma) \frac{2(m-k+1)}{3m}$. Hence, the total surfaces created in the construction can be estimated by $C(\sin(\gamma)m+1)$ (Color figure online)

1. Let $x \in (\frac{2 \sin(\gamma)}{\sqrt{3}}, 1)$. Then we define

$$V_K^{(3)}(x, y) = \begin{cases} \begin{pmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \\ 0 \\ 1 \end{pmatrix} & \text{if } \frac{\sin(\gamma)}{3} + \frac{1}{\sqrt{3}} \left(x - \frac{2 \sin(\gamma)}{\sqrt{3}} \right) \leq y, \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \text{else.} \end{cases}$$

2. Let $0 \leq k \leq K$ and $l \in \{0, \dots, m^k - 1\}$. Then we define for

$$(x, y) \in \left(-\frac{l}{m^k} \cos(\gamma), \frac{l}{m^k} \sin(\gamma) \right) + \frac{1}{m^k} B_\gamma^{(3)}$$

the function $V_K^{(3)}$ as

$$V_K^{(3)}(x, y) = V^{(3)} \left(m^k \left(x + \frac{l}{m^k} \cos(\gamma) \right), m^k \left(y - \frac{l}{m^k} \sin(\gamma) \right) \right).$$

3. Let $l \in \{0, \dots, m^{K+1} - 1\}$. We define for

$$(x, y) \in \text{conv} \left\{ \begin{pmatrix} -\frac{l}{m^{K+1}} \cos(\gamma) \\ \frac{l}{m^{K+1}} \sin(\gamma) \end{pmatrix}, \begin{pmatrix} -\frac{l}{m^{K+1}} \cos(\gamma) \\ \frac{l+1}{m^{K+1}} \sin(\gamma) \end{pmatrix}, \begin{pmatrix} -\frac{l+1}{m^{K+1}} \cos(\gamma) \\ \frac{l+1}{m^{K+1}} \sin(\gamma) \end{pmatrix} \right\}$$

the function $V_K^{(4)}$ as

$$V_K^{(3)}(x, y) = \begin{cases} \begin{pmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{pmatrix} & \text{if } y \geq \frac{l}{m^{K+1}} \sin(\gamma) + \frac{1}{\sqrt{3}} \left(x + \frac{l}{m^{K+1}} \cos(\gamma) - \frac{\sin(\gamma)}{\sqrt{3}m^{K+1}} \right) \\ & \text{and } y \leq \frac{l+2/3}{m^{K+1}} \sin(\gamma) + \frac{1}{\sqrt{3}} \left(x + \frac{l}{m^{K+1}} \cos(\gamma) - \frac{\sin(\gamma)}{\sqrt{3}m^{K+1}} \right) \\ & \text{and } x \geq -\frac{l}{m^{K+1}} \cos(\gamma) + \frac{\sin(\gamma)}{\sqrt{3}m^{K+1}}, \\ \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{pmatrix} & \text{if } y \geq \frac{l+1/3}{m^{K+1}} \sin(\gamma) - \frac{1}{\sqrt{3}} \left(x + \frac{l}{m^{K+1}} \cos(\gamma) \right) \\ & \text{and } y \leq \frac{l+1}{m^{K+1}} \sin(\gamma) - \frac{1}{\sqrt{3}} \left(x + \frac{l}{m^{K+1}} \cos(\gamma) \right) \\ & \text{and } x \leq -\frac{l}{m^{K+1}} \cos(\gamma) + \frac{\sin(\gamma)}{\sqrt{3}m^{K+1}}, \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \text{else.} \end{cases}.$$

4. Let $l \in \{0, \dots, m^{K+1} - 1\}$. We define for

$$(x, y) \in \text{conv} \left\{ \begin{pmatrix} -\frac{l}{m^{K+1}} \cos(\gamma) \\ \frac{l}{m^{K+1}} \sin(\gamma) \end{pmatrix}, \begin{pmatrix} -\frac{l}{m^{K+1}} \cos(\gamma) \\ \frac{l+1}{m^{K+1}} \sin(\gamma) \end{pmatrix}, \begin{pmatrix} -\frac{l+1}{m^{K+1}} \cos(\gamma) \\ \frac{l+1}{m^{K+1}} \sin(\gamma) \end{pmatrix} \right\}$$

the function $V_K^{(3)}$ as

$$V_K^{(3)}(x, y) = V_{\partial}^{(3)} \left(m^{K+1} \left(x + \frac{l}{m^{K+1}} \cos(\gamma) \right), m^{K+1} \left(y - \frac{l}{m^{K+1}} \sin(\gamma) \right) \right).$$

As in the case $N = 4$, we observe that $V_K^{(3)}$ is curl-free. Let $u_K^{(3)} : S_{\gamma} \rightarrow \mathbb{R}$ be a corresponding primitive with $u_K^{(3)}(0, 0) = 0$. As for $N = 4$, it holds by construction that $u_K(-t \cos(\gamma), t \sin(\gamma)) = 0$ for all $t \in (0, 1)$ and $u_K(t, \sin(\gamma)) = 0$ for all $t \in (-\cos(\gamma), 1)$. Next, we estimate similarly to before

$$\begin{aligned} \int_{S_{\gamma}} \text{dist} \left(\nabla u_K^{(3)}, K_{\pi/2,3} \right)^2 dx + \sigma |D^2 u_K^{(3)}| (S_{\gamma}) &\leq C \frac{1}{m^K} + C \sigma (K + \sin(\gamma) + 1) \\ &\leq C \left(\sin(\gamma)^K + \sigma K \right). \end{aligned}$$

Again, choosing $K = \left\lceil \frac{|\log \sigma|}{|\log \sin(\gamma)|} \right\rceil$ yields the estimate

$$\begin{aligned} E_{\sigma, \pi/2,3}(u_K; S_{\gamma}) &= \int_{S_{\gamma}} \text{dist} \left(\nabla u_K^{(3)}, K_{\pi/2,3} \right)^2 dx + \sigma |D^2 u_K^{(3)}| (S_{\gamma}) \\ &\leq C \sigma \left(\frac{|\log \sigma|}{|\log \sin(\gamma)|} + 1 \right). \end{aligned}$$

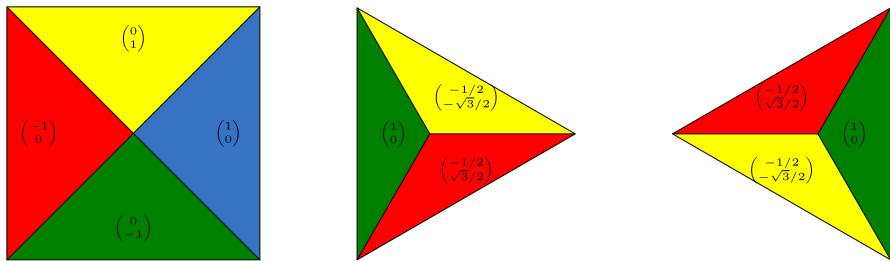


Fig. 10 Sketch of the functions constructed in Lemma 9 for Q_0 (left), $T_0^{(1)}$ (middle) and $T_0^{(2)}$ (right)

Step 3: Branching construction on $(0, 1)^2$ for $N = 3, 4$. We define $u : (0, 1)^2 \rightarrow \mathbb{R}$ as (c.f. also Fig. 3)

$$u(x, y) = \begin{cases} e^{i\gamma} \cdot (x, y - 1) & \text{if } e^{i\gamma} \cdot (x, y - 1) \geq 0, \\ u^{(N)}(e^{-i(\gamma - \pi/2)}(x, y)) & \text{if } e^{i\gamma} \cdot (x, y - 1) \leq 0, \end{cases}$$

where $u^{(N)}$ is the function constructed in Step 1 and Step 2 for $N = 3$ or $N = 4$, respectively. Then we conclude by Step 1c and 2c, respectively, that

$$\begin{aligned} E_{\sigma, \gamma, N}(u) &\leq E_{\sigma, \pi/2, N}(u^{(N)}) + C\sigma \\ &\leq C\sigma \left(\frac{|\log \sigma|}{|\log \sin(\gamma)|} + 1 \right) + C\sigma \leq C\sigma \left(\frac{|\log \sigma|}{|\log \sin(\gamma)|} + 1 \right). \end{aligned}$$

□

7 An Upper Bound by Covering and Solutions to the Differential Inclusion

In this section, we discuss how upper bounds for the energy $E_{\sigma, \gamma, N}(\cdot, \Omega)$ for $N = 3, 4$ and a bounded Lipschitz domain $\Omega \subseteq \mathbb{R}^2$ can be shown using a good covering of Ω with specific building blocks that allow for simple functions which only use the preferred gradients and satisfy zero boundary conditions. Iterating this construction leads to solutions of the differential inclusion $\nabla u \in K_{\gamma, N}$ and $u = 0$ on $\partial\Omega$ whose regularity can be controlled through interpolation. We start by constructing the building blocks.

Lemma 9 1. Let $Q_\gamma = e^{i\gamma}(-1/2, 1/2)^2$ be the unit cube centered at 0 and with two sides parallel to $e^{i\gamma}$. Then there exists a function $u \in W_0^{1, \infty}(Q_\gamma)$ such that $\nabla u \in BV(Q_\gamma; K_{\gamma, 4})$ and $|D^2 u|(Q_\gamma) \leq 4$.

2. Consider the triangles $T_\gamma^{(1)} = \text{conv} \left\{ e^{i\gamma}, e^{i\gamma + i\frac{2\pi}{3}}, e^{i\gamma + i\frac{4\pi}{3}} \right\}$ and $T_\gamma^{(2)} = \text{conv} \left\{ -e^{i\gamma}, -e^{i\gamma + i\frac{2\pi}{3}}, -e^{i\gamma + i\frac{4\pi}{3}} \right\}$. Then there exist functions $u^{(i)} \in W_0^{1, \infty}(T_\gamma^{(i)})$ such that $\nabla u^{(i)} \in BV(T_\gamma^{(i)}, K_{\gamma, N})$ and $|D^2 u^{(i)}| = 3\sqrt{3}$.

Proof For a visualization of the constructions for $\gamma = 0$, see Fig. 10.

For 1., note that by a change of coordinates it is enough to construct a function for $\gamma = 0$ and $Q_0 = (-1/2, 1/2)^2$. Then define $u : (-1/2, 1/2)^2 \rightarrow \mathbb{R}$ as

$$u(x, y) = \begin{cases} x - \frac{1}{2}, & \text{if } x \geq 0, -x \leq y \leq x, \\ -x - \frac{1}{2}, & \text{if } x \leq 0, x \leq y \leq -x, \\ y - \frac{1}{2}, & \text{if } y \geq 0, -y \leq x \leq y, \\ -y - \frac{1}{2}, & \text{if } y \leq 0, y \leq x \leq -y. \end{cases}$$

The claimed properties follow directly.

For 2., it suffices again to consider the case $\gamma = 0$. Then we define

$$u^{(1)}(x, y) = \begin{cases} x + \frac{1}{2} & \text{if } (x, y) \in \text{conv} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix} \right\}, \\ -\frac{1}{2}x - \frac{\sqrt{3}}{2}y + \frac{1}{2} & \text{if } (x, y) \in \text{conv} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \right\}, \\ -\frac{1}{2}x + \frac{\sqrt{3}}{2}y + \frac{1}{2} & \text{if } (x, y) \in \text{conv} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix} \right\} \end{cases}$$

and

$$u^{(2)}(x, y) = -u^{(1)}(-x, y).$$

All claimed properties can be immediately verified. $\square$

Next, we discuss a way of covering $\Omega \subseteq \mathbb{R}^2$ that will allow for simple construction of test functions using the functions from the previous lemma.

Definition 1 Let $\Omega \subseteq \mathbb{R}^2$, $N = 3, 4$, $\gamma \in (-\pi, \pi)$ and $\theta \in (0, 1)$. We say that a sequence of families $\mathcal{F}_K \subseteq \mathcal{P}(\Omega)$ is a (N, γ, θ) -covering of Ω if there exists $C_{N, \gamma, \theta} > 0$ such that the following properties are satisfied:

1. the family $\mathcal{F}_K$ consists of pairwise disjoint sets;
2. $\mathcal{F}_K \subseteq \mathcal{F}_{K+1}$ for every $K \in \mathbb{N}$;
3. (a) If $N = 3$: if $S \in \mathcal{F}_K$ then $S = a + \lambda T_\gamma^{(i)} \subseteq \Omega$ for some $i = 1, 2$, $a \in \mathbb{R}^2$ and $\lambda > 0$;
(b) If $N = 4$: if $S \in \mathcal{F}_K$ then $S = a + \lambda Q_\gamma \subseteq \Omega$ for some $a \in \mathbb{R}^2$ and $\lambda > 0$;
4. for every $K \in \mathbb{N}$ it holds $\mathcal{L}^2(\Omega \setminus \bigcup_{S \in \mathcal{F}_K} S) \leq C_{N, \gamma, \theta} \theta^K$;
5. for every $K \in \mathbb{N}$ it holds $\sum_{a + \lambda S \in \mathcal{F}_K} \lambda \leq C_{N, \gamma, \theta} K$.

If a domain Ω allows for a covering of the above type, this leads to good competitors for the energy $E_{\sigma, \gamma, N}(\cdot; \Omega)$ and solutions to the differential inclusion.

Proposition 10 Let $\Omega \subseteq \mathbb{R}^2$ be open, $N = 3, 4$. Let $\gamma \in \Gamma_N$ and $\theta \in (0, 1)$ such that Ω possesses a (N, γ, θ) -cover.

1. Then it holds

$$\min_{u \in \mathcal{A}_0(\Omega)} E_{\sigma, \gamma, N}(u; \Omega) \leq C C_{N, \gamma, \theta} \sigma \left(\frac{|\log \sigma|}{|\log \theta|} + 1 \right),$$

where $C > 0$ is a universal constant independent from θ and γ .

2. Then there exists a function $u \in W_0^{1, \infty}(\Omega)$ such that $\nabla u \in K_{\gamma, N}$ a.e. and $\nabla u \in W^{s, q}(\Omega)$ for all $0 < s < 1$, $q \in (1, \infty)$ such that $\frac{1}{q} > s$.

Proof Let $(\mathcal{F}_K)_K$ be a (N, γ, θ) -cover for Ω . Let $K \in \mathbb{N}$. If $N = 3$, then define the function $u_K^{(3)} : (0, 1)^2 \rightarrow \mathbb{R}$ as

$$u_K^{(3)}(x, y) = \begin{cases} \lambda u^{(i)}\left(\frac{(x, y) - a}{\lambda}\right) & \text{if } (x, y) \in a + \lambda T_\gamma^{(i)} \text{ for } i \in \{1, 2\}, a + \lambda T_\gamma^{(i)} \in \mathcal{F}_K, \\ 0 & \text{if } (x, y) \in (0, 1)^2 \setminus \bigcup_{S \in \mathcal{F}_K} S, \end{cases}$$

where $u^{(i)} : T_\gamma^{(i)} \rightarrow \mathbb{R}$ are the functions from Lemma 9 2. If $N = 4$ then define the function $u_K^{(4)} : (0, 1)^2 \rightarrow \mathbb{R}$ as

$$u_K^{(4)}(x, y) = \begin{cases} \lambda u\left(\frac{(x, y) - a}{\lambda}\right) & \text{if } (x, y) \in a + \lambda Q_\gamma \text{ for } a + \lambda Q_\gamma \in \mathcal{F}_K, \\ 0 & \text{if } (x, y) \in (0, 1)^2 \setminus \bigcup_{S \in \mathcal{F}_K} S, \end{cases}$$

where $u : Q_\gamma \rightarrow \mathbb{R}$ is the function from lemma 9 1.

Then $u_K^{(N)} \in \mathcal{A}_0$. Moreover, it holds by the properties of $\mathcal{F}_K$ and u (cf. Lemma 9) for $N = 4$ that

$$\begin{aligned} E_{\sigma, \gamma, N}(u_K^{(4)}) &\leq \mathcal{L}^2\left(\Omega \setminus \bigcup_{S \in \mathcal{F}_K} S\right) + \sigma \sum_{a + \lambda \cdot Q_\gamma \in \mathcal{F}_K} \left(\lambda |D^2 u|(Q_\gamma) + 2\lambda \mathcal{H}^1(\partial Q_\gamma)\right) \\ &\leq C C_{4, \gamma, \theta} \left(\theta^K + \sigma K\right). \end{aligned}$$

Then choose $K = \lceil \frac{|\log \sigma|}{|\log \theta|} \rceil \in [\frac{|\log \sigma|}{|\log \theta|}, \frac{|\log \sigma|}{|\log \theta|} + 1]$ which yields the desired upper bound. A similar calculation shows the upper bound for $N = 3$. This shows 1. For 2. we note that as in Ginster and Zwicknagl (2023, Proposition 12), we can use an interpolation argument (c.f. also Rüländ et al., 2018, Corollary 2.1) to show that the sequence of functions $u_K^{(N)}$ as constructed above has a limit $u^N \in W_0^{1, \infty}(\Omega)$ with the claimed properties. $\square$

Next, we show that every Lipschitz domain allows for a $(N, \gamma, \frac{1}{2})$ -cover for $N = 3, 4$ and $\gamma \in (-\pi, \pi)$.

Lemma 11 Let $N = 3, 4$, $\gamma \in (-\pi, \pi)$ and $\Omega \subseteq \mathbb{R}^2$ an open, bounded set with Lipschitz boundary. Then there exists a $(N, \gamma, \frac{1}{2})$ -cover of Ω . Moreover, the constants $C_{N, \gamma, 1/2}$ are uniformly bounded.

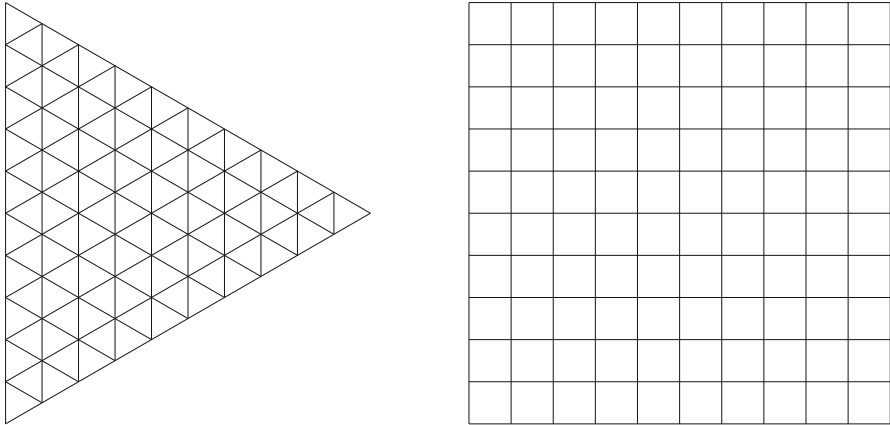


Fig. 11 Sketch of the lattices $\mathcal{L}^{(3)}$ and $\mathcal{L}^{(4)}$ for $\gamma = 0$

Proof Consider the lattices $\mathcal{L}^{(3)} = \{l_1 e^{i\gamma + i\pi/2} + l_2 e^{i\gamma + i\pi/6} : l_1, l_2 \in \mathbb{Z}\}$ and $\mathcal{L}^{(4)} = \{l_1 e^{i\gamma} + l_2 e^{i\gamma + i\pi/2} : l_1, l_2 \in \mathbb{Z}\}$, see Fig. 11. We define the families $\mathcal{F}_K^{(N)}$ inductively. Let

$$\begin{aligned} \mathcal{F}_1^{(3)} &= \left\{ \text{int}(\text{conv}\{i, j, k\}) : i, j, k \in \mathcal{L}^{(3)}, |i - j| = |i - k| = |j - k| = 1 \right. \\ &\quad \left. \text{and } \text{int}(\text{conv}\{i, j, k\}) \subseteq \Omega \right\}, \\ \mathcal{F}_1^{(4)} &= \left\{ \text{int}(\text{conv}\{h, i, j, k\}) : h, i, j, k \in \mathcal{L}^{(4)}, |i - j| = |j - k| \right. \\ &\quad \left. = |k - h| = |h - i| = 1 \right. \\ &\quad \left. \text{and } \text{conv}\{h, i, j, k\} \subseteq \Omega \right\}. \end{aligned}$$

Assume that $\mathcal{F}_K^{(3)}$ and $\mathcal{F}_K^{(4)}$ are already defined and set

$$\begin{aligned} S_{K+1}^{(3)} &= \{ \text{int}(\text{conv}\{i, j, k\}) : i, j, k \in 2^{-K} \mathcal{L}^{(3)}, |i - j| = |i - k| = |j - k| = 2^{-K}, \\ &\quad \text{int}(\text{conv}\{i, j, k\}) \subseteq \Omega \text{ and } \text{int}(\text{conv}\{i, j, k\}) \cap T = \emptyset \text{ for all } T \in \mathcal{F}_K^{(3)} \} \end{aligned}$$

and

$$\begin{aligned} S_{K+1}^{(4)} &= \{ \text{int}(\text{conv}\{h, i, j, k\}) : h, i, j, k \in 2^{-K} \mathcal{L}^{(4)}, \text{int}(\text{conv}\{h, i, j, k\}) \subseteq \Omega \\ &\quad |i - j| = |j - k| = |k - h| = |h - i| = 2^{-K} \text{ and} \\ &\quad \text{int}(\text{conv}\{i, j, k\}) \cap T = \emptyset \text{ for all } T \in \mathcal{F}_K^{(3)} \} \end{aligned}$$

Then define

$$\mathcal{F}_{K+1}^{(3)} = \mathcal{F}_K^{(3)} \cup S_{K+1}^{(3)} \quad \text{and} \quad \mathcal{F}_{K+1}^{(4)} = \mathcal{F}_K^{(4)} \cup S_{K+1}^{(4)}.$$

By definition, these families only consist of scaled and translated versions of the triangles $T_\gamma^{(i)}$ for $N = 3$ and the rotated square Q_γ for $N = 4$, respectively. It follows for a universal $C > 0$ that for $(x, y) \notin \bigcup_{S \in \mathcal{F}_K^{(N)}} S$ it holds $(x, y) \in B_{C2^{-K}}(\partial\Omega)$, where $B_{C2^{-K}}(\partial\Omega)$ denotes the set of all points with a distance of less than $C2^{-K}$ to the boundary of Ω . By the regularity of $\partial\Omega$, we find that

$$\mathcal{L}^2(\Omega \setminus \bigcup_{S \in \mathcal{F}_K^{(N)}} S) \leq C2^{-K}. \quad (15)$$

This implies also that $\#\mathcal{F}_{K+1}^{(N)} \setminus \mathcal{F}_K^{(N)} \leq C2^K$. In particular,

$$\sum_{a+\lambda S \in \mathcal{F}_K^{(N)}} \lambda \leq C\#\mathcal{F}_1^{(N)} + C \sum_{k=1}^{K-1} \#\mathcal{F}_{k+1}^{(N)} \setminus \mathcal{F}_k^{(N)} \cdot 2^{-k} \leq CK. \quad (16)$$

Note that the constant $C > 0$ in (15) and (16) can be chosen independently from γ . $\square$

As a direct consequence of Lemma 11 and Proposition 10, we find the following corollary.

Corollary 12 *Let $N = 3, 4$ and $\Omega \subseteq \mathbb{R}^2$ be an open bounded set with Lipschitz boundary. Then the following hold:*

1. *There exists a constant $C > 0$ (depending on Ω) such that*

$$\min_{u \in \mathcal{A}_0(\Omega)} E_{\sigma, \gamma, N}(u; \Omega) \leq C\sigma (|\log \sigma| + 1).$$

2. *There exists $u \in W_0^{1, \infty}(\Omega)$ such that $\nabla u \in BV_{loc}(\Omega; K_{\gamma, N})$ and $\nabla u \in W^{s, q}$ for all $0 < s < 1$, $q \in (0, \infty)$ satisfying $\frac{1}{q} > s$.*

Proof The only property that does not follow directly from Lemma 11 and Proposition 10 is the fact that it holds for the solution $u \in W_0^{1, \infty}(\Omega)$ of the differential inclusion constructed in Proposition 10 based on the covering from Lemma 11 that $\nabla u \in BV_{loc}(\Omega)$. However, in the proof of Lemma 11 it can be seen that for all $U \subset\subset \Omega$ there exists $K \in \mathbb{N}$ such that $U \subset \bigcup_{S \in \mathcal{F}_K} S$. In particular, it follows from the associated construction in the proof of Proposition 10 that $|D^2u|(U) \leq CC_{N, \gamma, 1/2}K$ i.e., $\nabla u \in BV_{loc}(\Omega)$. $\square$

However, clearly the result above is not optimal in terms of the minimal scaling with respect to the angle γ . Next, we show that for $N = 4$ the unit square allows for a much better covering. Precisely, $(0, 1)^2$ has a $(4, \gamma, |\sin(\gamma)|)$ -cover with uniformly bounded constants.

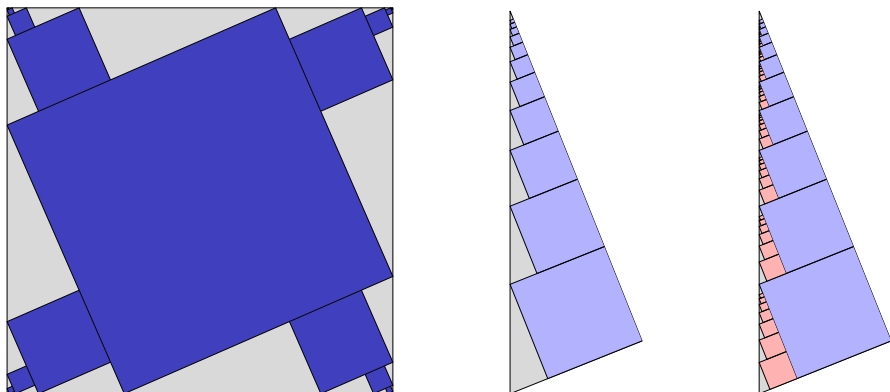


Fig. 12 Sketch of the covering of $(0, 1)^2$ described in Lemma 13. Left: the set $\mathcal{F}_1$ (blue). Middle: the set $\mathcal{F}_1$ (lightblue). Right: the set $\mathcal{F}_2$ (red) (Color figure online)

Lemma 13 *For every $\gamma \in (-\pi/4, \pi/4)$, the square $(0, 1)^2$ possesses a $(4, \gamma, |\sin(\gamma)|)$ -cover. Moreover, the constant $C_{4, \gamma, |\sin \gamma|}$ can be uniformly bounded.*

Proof Let us assume that $\gamma > 0$. For a visualization of the construction, see Fig. 12. For $\ell \in \mathbb{N}$, let $r_\ell = \frac{\sin(\gamma)}{\cos(\gamma) + \sin(\gamma)} \frac{\cos(\gamma)}{1 + \sin(\gamma) \cos(\gamma)} \left(\frac{\cos(\gamma) \sin(\gamma)}{1 + \cos(\gamma) \sin(\gamma)} \right)^\ell$. Then define the families

$$\begin{aligned} \mathcal{F}_1^{(1)} &= \left\{ \left(\frac{r_\ell}{2} (\cos(\gamma) + \sin(\gamma)) \right) + r_\ell \mathcal{Q}_\gamma : \ell \in \mathbb{N} \right\}, \\ \mathcal{F}_1^{(2)} &= \left\{ \left(1 - \frac{r_\ell}{2} (\cos(\gamma) + \sin(\gamma)) \right) + r_\ell \mathcal{Q}_\gamma : \ell \in \mathbb{N} \right\}, \\ \mathcal{F}_1^{(3)} &= \left\{ \left(1 - \frac{r_\ell}{2} (\cos(\gamma) + \sin(\gamma)) \right) + r_\ell \mathcal{Q}_\gamma : \ell \in \mathbb{N} \right\}, \\ \mathcal{F}_1^{(4)} &= \left\{ \left(\frac{r_\ell}{2} (\cos(\gamma) + \sin(\gamma)) \right) + r_\ell \mathcal{Q}_\gamma : \ell \in \mathbb{N} \right\}. \end{aligned}$$

Eventually, we set

$$\mathcal{F}_1 = \bigcup_{i=1}^4 \mathcal{F}_1^{(i)} \cup \left(\left(\frac{1}{2} \right) + \frac{1}{\cos(\gamma) + \sin(\gamma)} \mathcal{Q}_\gamma \right).$$

Note that $\mathcal{F}_1$ consists of pairwise disjoint sets. Then one computes

$$\begin{aligned} \mathcal{L}^2((0, 1)^2 \setminus \bigcup_{Q \in \mathcal{F}_1} Q) &= 1 - \frac{1}{(\cos(\gamma) + \sin(\gamma))^2} - 4 \sum_{\ell=0}^{\infty} r_\ell^2 \\ &= \frac{2 \sin(\gamma) \cos(\gamma)}{1 + 2 \sin(\gamma) \cos(\gamma)} - 4 \frac{\sin(\gamma)^2 \cos(\gamma)^2}{(1 + 2 \sin(\gamma) \cos(\gamma))^2} \end{aligned}$$

$$= \frac{2 \sin(\gamma) \cos(\gamma)}{(1 + 2 \sin(\gamma) \cos(\gamma))^2} \leq 2 \sin(\gamma)$$

and

$$\sum_{a+\lambda Q \in \mathcal{F}^1} \lambda = \frac{1}{\cos(\gamma) + \sin(\gamma)} + 4 \sum_{\ell=0}^{\infty} r_{\ell} = \frac{1 + 4 \sin(\gamma) \cos(\gamma)}{\cos(\gamma) + \sin(\gamma)} \leq C.$$

This shows all the needed properties of $\mathcal{F}_1$. To construct $\mathcal{F}_K$, we notice that $(0, 1)^2 \setminus \bigcup_{Q \in \mathcal{F}_1} Q$ can be written as the disjoint union $\bigcup_{k \in \mathbb{N}} T_k$ where T_k is a rotated (by multiples of $\frac{\pi}{2}$), dilated and translated version of the triangle $T = \text{conv} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \sin(\gamma) \begin{pmatrix} \cos(\gamma) \\ \sin(\gamma) \end{pmatrix} \right\}$ such that $\sum_{k \in \mathbb{N}} \mathcal{H}^1(\partial T_k) \leq C$, see Fig. 12. In order to construct the families $\mathcal{F}_K$ for $K \geq 2$ inductively, it is then enough to show that there exists a disjoint family $\mathcal{F}$ of dilated and translated versions of Q_{γ} such that $T \setminus \bigcup_{a+\lambda Q_{\gamma} \in \mathcal{F}} a + \lambda Q_{\gamma}$ can be written as the disjoint union of translated and dilated versions of T such that (see Fig. 12)

$$\sum_{a+\lambda Q_{\gamma} \in \mathcal{F}} \lambda \leq 1 \text{ and } \mathcal{L}^2(T \setminus \bigcup_{a+\lambda Q_{\gamma} \in \mathcal{F}} a + \lambda Q_{\gamma}) \leq \sin(\gamma) \mathcal{L}^2(T).$$

Then one can inductively define families $\tilde{\mathcal{F}}_K$ for T such that (cf. Fig. 12)

$$\tilde{\mathcal{F}}_K \subseteq \tilde{\mathcal{F}}_{K-1}, \quad \sum_{a+\lambda Q_{\gamma} \in \tilde{\mathcal{F}}_K} \lambda \leq K \text{ and } \mathcal{L}^2(T \setminus \bigcup_{a+\lambda Q_{\gamma} \in \tilde{\mathcal{F}}_K} a + \lambda Q_{\gamma}) \leq \sin(\gamma)^K \mathcal{L}^2(T).$$

Eventually, the families $\mathcal{F}_K$ are simply given by $\mathcal{F}_1$ together with the corresponding families $\tilde{\mathcal{F}}_{K-1}$ for the triangles T_k .

Hence, it remains to show the existence of the family $\mathcal{F}$. For $\ell \in \mathbb{N}$, we define $\tilde{r}_{\ell} = \frac{\sin(\gamma) \cos(\gamma)}{\sin(\gamma) + \cos(\gamma)} \left(\frac{\cos(\gamma)}{\sin(\gamma) + \cos(\gamma)} \right)^{\ell}$ and $\delta_{\ell} = \left(\frac{\cos(\gamma)}{\sin(\gamma) + \cos(\gamma)} \right)^{\ell}$. Then set (cf. Fig. 12)

$$\mathcal{F} = \left\{ \left(\frac{\tilde{r}_{\ell} (\cos(\gamma) + \sin(\gamma))}{\frac{\tilde{r}_{\ell}}{2} (\cos(\gamma) + \sin(\gamma)) + \tilde{r}_{\ell} \frac{\sin(\gamma)^2}{\cos(\gamma)} + \sum_{k=0}^{\ell-1} \frac{\sin(\gamma)}{\sin(\gamma) + \cos(\gamma)} \delta_k} \right) + \tilde{r}_{\ell} Q_{\gamma} : \ell \in \mathbb{N} \right\}.$$

It follows that $\mathcal{F}$ is a pairwise disjoint family. Moreover, we compute (recall that $\cos(\gamma) \geq \cos(\pi/4) \geq 1/2$)

$$\begin{aligned} \mathcal{L}^2(T \setminus \bigcup_{Q \in \mathcal{F}} Q) &= \frac{\sin(\gamma) \cos(\gamma)}{2} - \sum_{k=0}^{\infty} \tilde{r}_{\ell}^2 \\ &= \frac{\sin(\gamma) \cos(\gamma)}{2} \left(1 - 2 \frac{\sin(\gamma) \cos(\gamma)}{\sin(\gamma)^2 + 2 \sin(\gamma) \cos(\gamma)} \right) \\ &\leq \mathcal{L}^2(T) \frac{\sin(\gamma)}{2 \cos(\gamma)} \leq \sin(\gamma) \mathcal{L}^2(T) \end{aligned}$$

and

$$\sum_{a+\lambda Q_\gamma \in \mathcal{F}} \lambda = \sum_{\ell=0}^{\infty} \tilde{r}_\ell = \cos(\gamma) \leq 1.$$

□

Remark 2 Similarly, it can be shown that there are families $\mathcal{F}_K$ consisting of dilated and translated versions of the building blocks $T_\gamma^{(1)}$ and $T_\gamma^{(2)}$ satisfying (1) - (3) when $(0, 1)^2$ is replaced by, for example, $T_0^{(1)}$ or $T_0^{(2)}$.

Combining the previous result and Lemma 13 allows us to prove a scaling law for $\min_{u \in \mathcal{A}_0((0,1)^2)} E_{\sigma,\gamma,4}(u)$, Theorem 2.

Proof of Thm. 2 Step 1: Upper bounds. Clearly, $u = 0$ satisfies $u \in \mathcal{A}_0$ and $E_{\sigma,\gamma,N}(u) = 1$. The other upper bounds follow directly from Corollary 12 for $N = 3$ and from Proposition 10 and Lemma 13 for $N = 4$.

Step 2: Lower bounds. Let us first consider $N = 3$. First, note that for $\gamma \in \Gamma_3$ with $|\gamma| \geq \pi/12$ the lower bound follows from Proposition 5. On the other hand, for $|\gamma| \leq \pi/12$ the lower bound follows from a similar argument and the fact that $u(\cdot, 0) = 0$ and for $\gamma \in \Gamma_3$ it holds for $\xi \in K_{\gamma,3}$ that $|\xi_1| \geq \min\{\cos(\gamma), |\cos(\gamma + 2\pi/3)|, |\cos(\gamma + 4\pi/3)|\} \geq |\cos(7\pi/12)| > 0$.

Next, let $N = 4$. First we claim that

$$\min_{u \in \mathcal{A}_0} E_{\sigma,\gamma,4}(u) \geq \frac{1}{6} \min\{1, \sigma\}. \quad (17)$$

Let $u \in \mathcal{A}_0$. We may assume that $E_{\sigma,\gamma,4}(u) \leq \frac{1}{6} \min\{1, \sigma\}$ (otherwise there is nothing to show). Now, find $\bar{x} \in (0, 1)$ and $\bar{y} \in (0, 1)$ such that

$$\begin{aligned} \int_0^1 \text{dist}(\nabla u(x, \bar{y}), K_{\gamma,4})^2 dx + \sigma |\partial_1 \nabla u(\cdot, \bar{y})| (0, 1) &\leq E_{\sigma,\gamma,4}(u) \\ \text{and } \int_0^1 \text{dist}(\nabla u(\bar{x}, y), K_{\gamma,4})^2 dy + \sigma |\partial_2 \nabla u(\bar{x}, \cdot)| (0, 1) &\leq E_{\sigma,\gamma,4}(u). \end{aligned} \quad (18)$$

As $\nabla u \in BV((0, 1)^2; \mathbb{R}^2)$ and $u = 0$ on $\partial(0, 1)^2$, we have in the sense of traces $\partial_2 u(0, y) = 0$ and $\partial_1 u(x, 0) = 0$. Using (18) we obtain the estimate $|\nabla u(x, \bar{y})| \leq 3/6 = 1/2$ for a.e. $x \in (0, 1)$. Hence, it follows that

$$\frac{1}{4} \leq \int_0^1 \text{dist}(\nabla u(x, \bar{y}), K_{\gamma,N})^2 dx \leq E_{\sigma,\gamma,N}(u).$$

This shows claim (17). Then the lower bound for $\sigma \geq 1$ follows immediately. Next, fix $K \in \mathbb{N}$ from Proposition 6 and notice that for $1 \geq \sigma \geq |\sin(\gamma)|^K$ it holds

$$\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \leq K + 1.$$

Consequently, it follows from estimate (17)

$$\min_{u \in \mathcal{A}_0} E_{\sigma, \gamma, 4}(u) \geq \frac{1}{6} \min\{1, \sigma\} \geq \frac{1}{6(K+1)} \min\left\{1, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right)\right\}.$$

Eventually, we note that the lower bound for $\sigma \leq |\sin(\gamma)|^K$ follows from Proposition 6. This finishes the proof for $N = 4$. $\square$

Remark 3 One can argue similarly to show that there exist $C, c > 0$ such that for $i = 1, 2$ and $\gamma \in \Gamma_3$, c.f. Remark 2,

$$\begin{aligned} & c \min\left\{1, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right)\right\} \\ & \leq \min_{u \in \mathcal{A}_0} E_{\sigma, \gamma, 3}(u; T_0^{(i)}) \leq C \min\left\{1, \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right)\right\}. \end{aligned}$$

Appendix

Here, we present the calculations that show Remark 1. Let us start with the case $N = 2$. For $\gamma \in \Gamma_2$, we note the following:

- if $\sigma \geq |\sin(\gamma)|^2$ then

$$\min\left\{\sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2\right\} = |\sin(\gamma)|^2;$$

- if $|\sin(\gamma)|^3 \leq \sigma \leq |\sin(\gamma)|^2$ then

$$\begin{aligned} \frac{1}{2}(\sigma + |\sin(\gamma)|^3) & \leq \sigma \leq \min\left\{\sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2\right\} \\ & \leq \sigma + |\sin(\gamma)|^3; \end{aligned}$$

Note that in this regime it holds $\sigma \leq \sigma + |\sin(\gamma)|^3 \leq 2\sigma$.

- if $\sigma \leq |\sin(\gamma)|^3$ then

$$\begin{aligned} \frac{1}{2}(\sigma^{2/3} |\sin(\gamma)| + \sigma) & \leq \sigma^{2/3} |\sin(\gamma)| \\ & \leq \min\left\{\sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2\right\} \\ & \leq \sigma^{2/3} |\sin(\gamma)| + \sigma. \end{aligned}$$

Hence, we find that

$$\min \left\{ \sigma^{2/3} |\sin(\gamma)| + \sigma, \sigma + |\sin(\gamma)|^3, |\sin(\gamma)|^2 \right\} \\ \sim \begin{cases} |\sin(\gamma)|^2 & \text{if } \sigma \geq |\sin(\gamma)|^2, \\ \sigma & \text{if } |\sin(\gamma)|^2 \geq \sigma \geq |\sin(\gamma)|^3, \\ |\sin(\gamma)| \sigma^{2/3} & \text{if } |\sin(\gamma)|^3 \geq \sigma. \end{cases}$$

For $N = 3, 4$ and $\gamma \in \Gamma_N$ we find the following:

- if $\sigma \geq |\sin(\gamma)|^2$ then

$$|\sin(\gamma)|^2 = \min \left\{ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right), |\sin(\gamma)|^2 \right\}.$$

- if $\sigma \leq |\sin(\gamma)|^2$ then

$$\begin{aligned} \frac{|\log \sigma|}{|\log |\sin(\gamma)||} - 2 &\leq \frac{|\log (\sigma/|\sin(\gamma)|^2)|}{|\log |\sin(\gamma)||} \\ &\leq \frac{|\sin(\gamma)|^2}{\sigma |\log |\sin(\gamma)||} \\ &\leq \frac{|\sin(\gamma)|^2}{\sigma |\log |\sin(\pi/4)||} = 2 \frac{|\sin(\gamma)|^2}{\sigma \log 2}, \end{aligned}$$

since $|\log |\sin(\pi/4)|| = \log \sqrt{2} = \log(2)/2$. Hence, it follows

$$\frac{\log(2)}{4} \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right) \leq \frac{1}{2} |\sin(\gamma)|^2 + \frac{3 \log(2)}{4} \sigma \leq |\sin(\gamma)|^2$$

and consequently,

$$\min \left\{ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right), |\sin(\gamma)|^2 \right\} \geq \frac{\log(2)}{4} \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right).$$

In particular, we have

$$\min \left\{ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right), |\sin(\gamma)|^2 \right\} \sim \begin{cases} |\sin(\gamma)|^2 & \text{if } \sigma \geq |\sin(\gamma)|^2, \\ \sigma \left(\frac{|\log \sigma|}{|\log |\sin(\gamma)||} + 1 \right) & \text{if } \sigma \leq |\sin(\gamma)|^2. \end{cases}$$

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Declarations

Competing interests The authors declare no competing interests.

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